KTU 2024 SCHEME · PCECT303 · S3 ECE

Analog Circuits Playbook

From "what does a capacitor do to a square wave?" to oscillators and power supplies — one friendly step at a time.

INSIDE

M1	RC wave shaping, clippers & clampers, BJT and MOSFET biasing
M2	BJT & MOSFET amplifiers — hybrid-π, frequency response, Miller, multistage, cascode
M3	Feedback amplifiers and oscillators — RC, Wien, Hartley, Colpitts, crystal
M4	Power amplifiers (A, B, AB, C) and linear voltage regulators
+	Placement & GATE rapid-fire · formula sheet · night-before summary

★ How to use this playbook

Analog circuits is where electronics stops being "just equations" and starts behaving like real hardware: waveforms get bent, transistors get parked at the right spot, gains roll off at high frequency, feedback tames everything, and power has to be delivered without burning the transistor. Every core-electronics interview — analog design, VLSI, embedded, test — dips into this subject.

Each topic follows the same rhythm: **plain-words intuition first** (as if you have never seen it before), then the **proper theory and derivation** at KTU end-sem depth, then **solved examples** that get gradually harder, and finally an **interview corner** with what companies and GATE like to ask. Read the intuition even if you "already know" the topic — that is usually where good interview answers come from.

Suggested routine: one section a day, try each example yourself before peeking, and use the module's quick-revision page the night before a test. Every numerical answer in this book was checked with a short Python calculation. Unless a question says otherwise we take V_{BE} = 0.7 V, V_{γ} = 0.7 V (silicon) and V_T (thermal voltage) = 25 mV.

The boxes you will see

Key idea — the one thing to remember

Think of it like — an everyday picture

Exam tip — how marks are won

Common mistake — how marks are lost

Interview corner — real placement-style Q&A

Formula — results worth memorising

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01

Wave Shaping & Biasing

> *RC shaping, clippers, clampers and a correctly placed Q-point are the first things any analog interviewer checks — and the first 25 marks of the paper.*

SYLLABUS CHECKLIST

- ☐ RC differentiator and integrator — response to step, pulse and square inputs
- ☐ First-order RC low-pass and high-pass filters
- ☐ Diode clippers — series, shunt, biased, double-ended
- ☐ Diode clampers — positive, negative, biased
- ☐ BJT biasing — DC load line, Q-point, fixed bias, collector-to-base bias, voltage-divider bias, stability factor
- ☐ MOSFET biasing — voltage-divider bias, self bias, drain-feedback bias

Module 1 · CO1 · Analyse wave-shaping circuits and design stable bias for BJT and MOSFET

1.1 RC circuits — the one-minute picture

Put a resistor and a capacitor together and you get the simplest "shaping" circuit in electronics. Everything about it is controlled by one number: the **time constant τ = RC** (in seconds, if R is in ohms and C in farads).

> **THINK OF IT LIKE**
>
> A capacitor is a bucket and the resistor is a narrow pipe feeding it. τ tells you how long the bucket takes to fill. A big pipe or a small bucket (small RC) fills almost instantly; a thin pipe or huge bucket (large RC) fills slowly. After 1τ it is 63 % full, after 5τ it is "full" (99.3 %).

There are only two ways to take the output, and they behave opposite to each other (Fig 1.1):

- **Output across R** (C in series) → **high-pass** circuit. Sudden changes pass through, steady (DC) levels are blocked. With a small RC it acts as a **differentiator**.
- **Output across C** (R in series) → **low-pass** circuit. Slow changes pass, fast wiggles are smoothed. With a large RC it acts as an **integrator**.

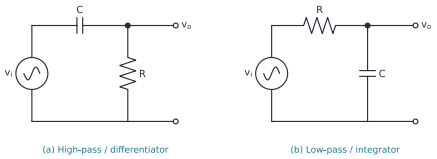


Fig 1.1 The two first-order RC circuits.

> **KEY IDEA**
>
> A capacitor's voltage cannot change instantly. So at the instant the input jumps, the jump appears **entirely across R**. That one fact explains every waveform in this section.

1.2 RC high-pass circuit and the differentiator

Step response

Apply a step of height V at t = 0. The capacitor starts empty, so the whole jump appears across R: $v_o(0^+) = V$. As C charges, current falls and the output decays to zero:

> **FORMULA**
>
> $$v_o(t) = V\, e^{-t/RC} \quad \text{(high-pass, step input)}$$

Pulse response

A pulse of width t_p is a step up at t = 0 followed by a step down at $t = t_p$. During the pulse the output decays from V to $V\,e^{-t_p/RC}$. At t_p the input falls by V, so the output also falls by V — to $V(e^{-t_p/RC} - 1)$, a negative value — and then decays back to zero (Fig 1.2). The droop on the top of the pulse is called **tilt**:

TILT

$$P = \frac{V - V\,e^{-t_p/RC}}{V} \times 100\,\% \approx \frac{t_p}{RC} \times 100\,\% \quad (\text{when } t_p \ll RC)$$

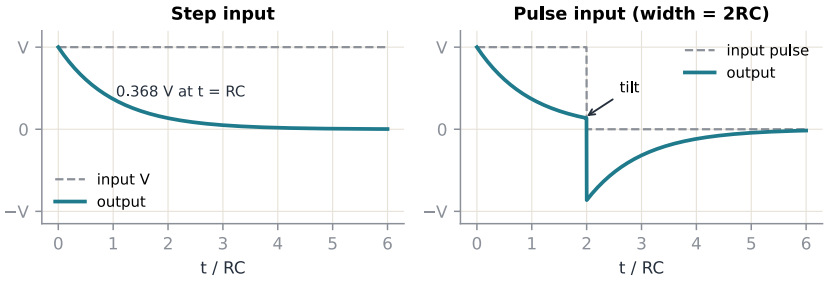


Fig 1.2 High-pass circuit: step response (left) and pulse response (right).

Square-wave response

A square wave is just a string of up and down steps. What you see depends on how RC compares with the period T (Fig 1.3, top row):

- **RC ≪ T**: the output decays fully within each half cycle → sharp positive and negative **spikes** at every edge. This is differentiation.
- **RC ≈ T/2**: partial decay → exponential "sawtooth-like" segments.
- **RC ≫ T**: hardly any decay → the square wave comes through with a small tilt (this is how a coupling capacitor should behave).

For a symmetric square wave of peak-to-peak V, in steady state the output alternates between V_1 and $V_1' = V_1 e^{-T/2RC}$ (and the negative mirror images). Because each edge is a jump of V, $V_1 + V_1' = V$, which gives

FORMULA

$$V_1 = \frac{V}{1 + e^{-T/2RC}} \qquad V_1' = V_1\,e^{-T/2RC}$$

$$\text{Tilt } P = \frac{V_1 - V_1'}{V/2} \times 100\,\% \approx \frac{T}{2RC} \times 100\,\% = \frac{\pi\,f_1}{f} \times 100\,\%, \quad f_1 = \frac{1}{2\pi RC}$$

Why it differentiates

The current is $i = C \cdot d(v_i - v_o)/dt$ and $v_o = iR$. If RC is very small, v_o stays tiny compared with v_i, so $v_i - v_o \approx v_i$:

DIFFERENTIATOR

$$v_o \approx RC\,\frac{dv_i}{dt} \qquad \text{condition: } RC \ll T \text{ (in practice } RC \le T/10)$$

So a ramp in gives a constant out, a square wave in gives spikes out, and a triangle wave in gives a square wave out.

SOLVED EXAMPLE 1.1 · KTU

A 1 kHz symmetric square wave of 10 V peak-to-peak is applied to a high-pass RC circuit with R = 10 kΩ and C = 0.1 µF. Find the steady-state output levels and the percentage tilt.

- RC = 10 k × 0.1 µ = 1 ms; T = 1 ms, so T/2 = 0.5 ms and T/2RC = 0.5.
- $e^{-0.5} = 0.6065 \rightarrow V_1 = 10/(1 + 0.6065) = 6.22$ V.
- $V_1' = 6.22 \times 0.6065 = 3.78$ V. Output swings +6.22 → +3.78, jumps to −6.22 → −3.78, and repeats.
- Tilt = (6.22 − 3.78)/5 × 100 = 49.0 % (the approximate formula T/2RC × 100 gives 50 %).

$V_1 = 6.22$ V, $V_1' = 3.78$ V, tilt ≈ 49 %

1.3 RC low-pass circuit and the integrator

Step and pulse response

Now the output is across C, which starts at zero and charges up:

FORMULA

$$v_o(t) = V\,(1 - e^{-t/RC}) \quad \text{(low-pass, step input)}$$

$$\text{Rise time (10 \% → 90 \%): } t_r = 2.2\,RC = \frac{0.35}{f_2}, \quad f_2 = \frac{1}{2\pi RC}$$

For a pulse, the output rises during t_p and decays after it. To pass a pulse with little distortion you need $t_r \ll t_p$, i.e. upper cut-off f_2 well above $1/t_p$. This rise-time–bandwidth link ($t_r \times f_2 = 0.35$) is a favourite interview question.

Square-wave response and the integrator

Fig 1.3 (bottom row): with **RC ≪ T** the output is a square wave with rounded corners; with **RC ≈ T/2** it is exponential up-and-down; with **RC ≫ T** it becomes a small **triangular wave** riding at the average level — the capacitor is integrating. Peak-to-peak of that triangle ≈ V·T/(4RC).

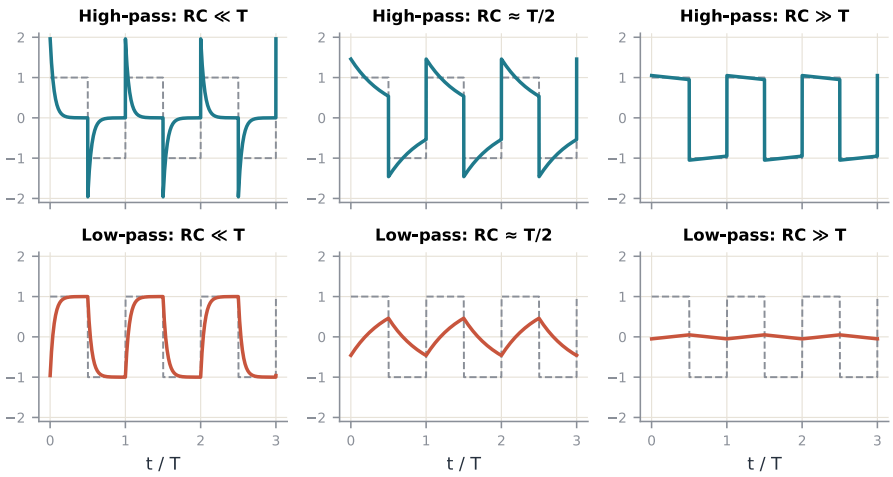


Fig 1.3 Square-wave response of high-pass (top) and low-pass (bottom) RC circuits for three values of RC/T. Dashed = input.

Why it integrates: $i = (v_i - v_o)/R$ and $v_o = (1/C)\int i\,dt$. If RC is large, v_o stays small, so $i \approx v_i/R$:

INTEGRATOR

$$v_o \approx \frac{1}{RC}\int v_i\,dt \qquad \text{condition: } RC \gg T \text{ (in practice } RC \geq 10T)$$

COMMON MISTAKE

Mixing up the conditions. **Differentiator → small RC** (output across R). **Integrator → large RC** (output across C). The price of a "good" integrator or differentiator is a very small output.

1.4 First-order RC low-pass and high-pass filters

The same two circuits, looked at with sine waves instead of steps, are **filters**. Use the voltage-divider rule with $Z_C = 1/(j\omega C)$:

TRANSFER FUNCTIONS

$$\text{Low-pass: } H(j\omega) = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\,f/f_c} \qquad |H| = \frac{1}{\sqrt{(1 + (f/f_c)^2)}},\quad \varphi = -\tan^{-1}(f/f_c)$$

$$\text{High-pass: } H(j\omega) = \frac{j\omega RC}{1 + j\omega RC} = \frac{1}{1 - j\,f_c/f} \qquad |H| = \frac{1}{\sqrt{(1 + (f_c/f)^2)}},\quad \varphi = +\tan^{-1}(f_c/f)$$

$$\text{Cut-off (−3 dB, half-power) frequency: } f_c = \frac{1}{2\pi RC}$$

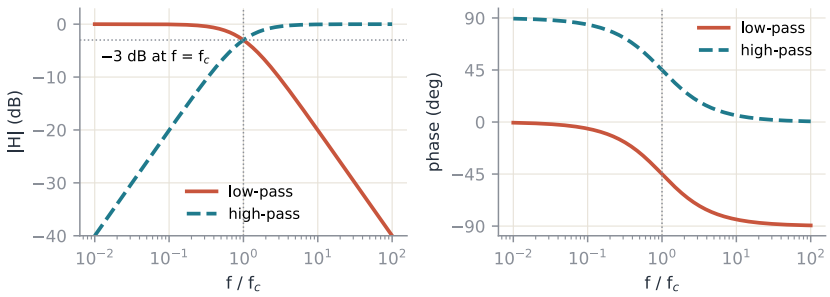


Fig 1.4 Bode magnitude and phase of first-order RC filters. Beyond f_c the low-pass falls at −20 dB/decade (−6 dB/octave).

KEY IDEA

At $f = f_c$: gain = $1/\sqrt{2} = 0.707$ (−3 dB), power is halved, and the phase is exactly halfway (−45° for low-pass, +45° for high-pass). One RC = one pole = 20 dB/decade.

SOLVED EXAMPLE 1.2 · FILTER

A low-pass RC filter uses R = 1 kΩ, C = 0.1 µF. Find f_c, and the gain (ratio and dB) and phase at 5 kHz.

- $f_c = 1/(2\pi \times 1000 \times 0.1 \times 10^{-6}) = 1591.5$ Hz ≈ 1.59 kHz.
- $f/f_c = 5000/1591.5 = 3.142 \rightarrow |H| = 1/\sqrt{(1 + 9.87)} = 0.303$.
- In dB: 20 log 0.303 = −10.4 dB. Phase = $-\tan^{-1}(3.142) = -72.3°$.

f_c = 1.59 kHz; at 5 kHz: 0.303 (−10.4 dB), −72.3°

INTERVIEW CORNER

Q. Why is a high-pass RC circuit called a differentiator? When does that fail?

A. Its output ≈ $RC \cdot dv_i/dt$ only if $RC \ll T$, so that $v_o \ll v_i$. If RC is comparable to T, the output is no longer the derivative — it is just a tilted copy.

Q. Relation between rise time and bandwidth?

A. For a single-pole system $t_r = 0.35/f_{3dB}$. Faster edges need more bandwidth — the reason scopes are spec'd by bandwidth.

Q. A coupling capacitor is really what kind of circuit?

A. A high-pass circuit with the next stage's input resistance. It must have $RC \gg T$ so that the signal passes with negligible tilt.

1.5 Diode clippers

A **clipper** (limiter) chops off the part of a waveform that goes above or below a chosen level, leaving the rest untouched. Think of a bouncer at a door with a height bar — anything taller gets cut to the bar's height.

We use the **constant-voltage-drop diode model**: the diode is open when reverse-biased and is a 0.7 V battery (V_γ) when it conducts. (If a question says "ideal diode", put $V_\gamma = 0$.)

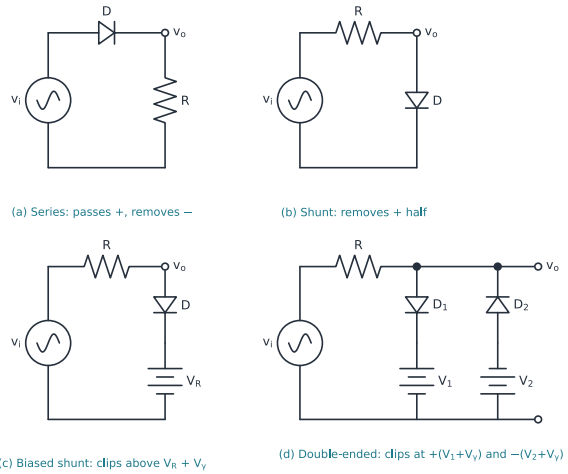


Fig 1.5 Clipper family: series, shunt, biased shunt and double-ended (two-level).

Series clipper (Fig 1.5a)

The diode is *in series* with the load. It conducts only on the positive half, so $v_o = v_i - V_\gamma$ when $v_i > V_\gamma$ and $v_o = 0$ otherwise. The negative half is removed (a half-wave rectifier is a series clipper!). Reverse the diode to remove the positive half instead.

Shunt clipper (Fig 1.5b)

The diode is *across* the output. When $v_i < V_\gamma$ the diode is off and $v_o = v_i$. When v_i tries to exceed V_γ the diode conducts and holds $v_o = V_\gamma$; the extra voltage drops across R.

Biased clipper (Fig 1.5c)

Add a battery V_R in series with the shunt diode and the clipping level moves to $\mathbf{V_R + V_\gamma}$. Flip the battery and you get clipping at $-V_R + V_\gamma$, etc.

Double-ended (two-level) clipper (Fig 1.5d)

Two biased diodes facing opposite ways. D_1 clips the top at $+(V_1 + V_\gamma)$; D_2 clips the bottom at $-(V_2 + V_\gamma)$. With a large sine input this produces a near-square wave — a classic way to make a "slicer".

> **EXAM TIP**
>
> Always do clippers in three steps: (1) find the input voltage at which each diode just turns on; (2) write v_o in each region; (3) draw the **transfer characteristic** (v_o vs v_i) and the output waveform one below the other. The transfer curve alone earns marks.

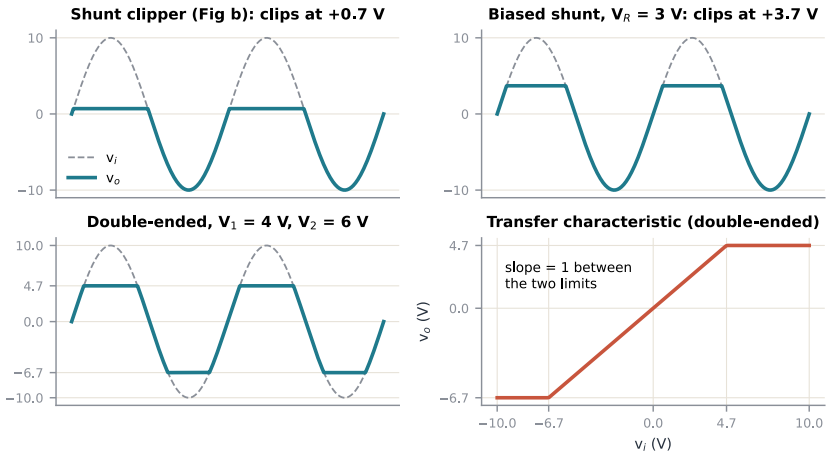


Fig 1.6 Outputs for $v_i = 10 \sin \omega t$ with $V_\gamma = 0.7$ V — the numbers of Example 1.3.

SOLVED EXAMPLE 1.3 · BIASED AND DOUBLE-ENDED CLIPPERS

$v_i = 10 \sin \omega t$, silicon diodes ($V_\gamma = 0.7$ V). Find the output of (a) the biased clipper of Fig 1.5c with $V_R = 3$ V, (b) the double-ended clipper of Fig 1.5d with $V_1 = 4$ V, $V_2 = 6$ V.

- (a) The diode turns on when v_o tries to exceed $V_R + V_\gamma = 3.7$ V. Below that $v_o = v_i$; above it $v_o = 3.7$ V. Negative half untouched (−10 V).
- (b) D_1 on when $v_o \geq 4.7$ V; D_2 on when $v_o \leq -6.7$ V. In between both are off and $v_o = v_i$ (slope 1).
- Transfer curve for (b): flat at −6.7 V, slope-1 line, flat at +4.7 V (Fig 1.6, bottom right).

(a) clipped at +3.7 V (b) limited between −6.7 V and +4.7 V

1.6 Diode clampers

A **clamper** does not change the shape at all — it **slides the whole waveform up or down** so that one of its peaks sits at a chosen level. It adds a DC level. (Clippers change shape; clampers change DC level.)

THINK OF IT LIKE

Lifting a whole bookshelf so its bottom shelf rests on a table. The shelves (shape) stay the same distance apart — only the height changes.

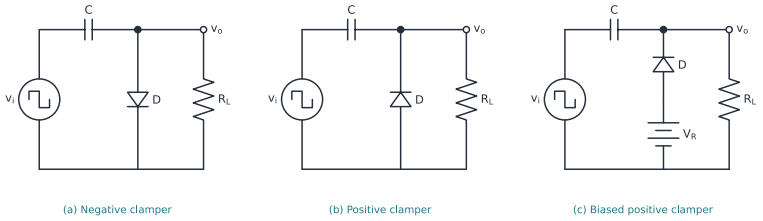


Fig 1.7 Negative, positive and biased clampers. A clamper needs a capacitor, a diode and a large load resistor.

How a negative clamper works (Fig 1.7a)

1. On the first positive half-cycle the diode conducts and C charges quickly (through the small diode resistance) to $V_m - V_\gamma$, with the input-side plate positive.
2. When the input swings negative the diode turns off. C has no fast discharge path (only the large R_L), so it keeps its voltage like a battery.
3. Output = $v_i - (V_m - V_\gamma)$: the waveform shifts down so that its positive peak sits at $+V_\gamma$ and its negative peak at $-(2V_m - V_\gamma)$.

Reverse the diode (Fig 1.7b) and the waveform is pushed *up*: negative peak at $-V_\gamma$, positive peak at $2V_m - V_\gamma$ — a **positive clamper**. Adding a battery V_R in series with the diode (Fig 1.7c) moves the clamping level to $V_R - V_\gamma$ (for the polarity shown).

DESIGN CONDITION

$R_L C \gg T/2$ (usually $5R_L C \geq 10 \times T/2$) — so C hardly discharges in a half-cycle

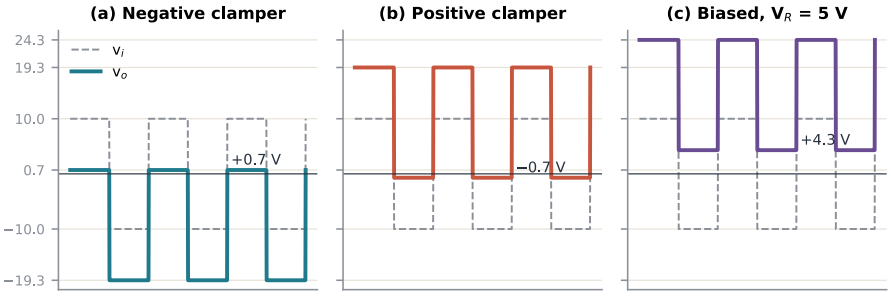


Fig 1.8 ±10 V square wave through the three clampers of Fig 1.7 (V_γ = 0.7 V, V_R = 5 V). Peak-to-peak stays 20 V every time.

SOLVED EXAMPLE 1.4 · CLAMPERS

A ±10 V square wave drives (a) a negative clamper, (b) a positive clamper and (c) a positive clamper biased with V_R = 5 V (Fig 1.7c). V_γ = 0.7 V and $R_L C \gg T$. Find the output levels.

▸ (a) C charges to 10 − 0.7 = 9.3 V. $v_o = v_i - 9.3$ → levels +0.7 V and −19.3 V.

▸ (b) C charges to 9.3 V the other way. $v_o = v_i + 9.3$ → levels −0.7 V and +19.3 V.

▸ (c) Diode conducts when v_o falls to $V_R - V_\gamma$ = 4.3 V, so the lowest level is clamped at 4.3 V → levels +4.3 V and +24.3 V.

▸ Check: peak-to-peak is 20 V in every case √.

(a) +0.7/−19.3 V (b) −0.7/+19.3 V (c) +4.3/+24.3 V

INTERVIEW CORNER

Q. Clipper vs clamper in one line?
A. A clipper changes the waveform's shape (removes a part); a clamper keeps the shape and changes only its DC level.

Q. What happens to a clamper if R_LC is too small?
A. C discharges during the half-cycle when the diode is off, so the output tilts and the clamp level wanders — the output is distorted.

Q. Where are clampers used?
A. DC restoration in TV video signals, voltage multipliers (a clamper + peak detector = voltage doubler), and level-shifting before an ADC.

1.7 BJT biasing — load line and Q-point

An amplifier transistor must be "parked" in the middle of its active region before any signal arrives, so that the signal can swing up and down without hitting saturation or cut-off. That parking spot is the **quiescent point (Q-point)**, given by (V_{CEQ}, I_{CQ}). **Biasing** is the DC circuit that sets it.

THINK OF IT LIKE

Parking a swing at its lowest point before you push it. Park it too close to one side and the first push makes it bang into the frame (clipping).

DC load line

KVL on the collector loop of a simple CE stage: $V_{CC} = I_CR_C + V_{CE}$. Plotted on the output characteristics this is a straight line, the **DC load line**, with end points

LOAD LINE END POINTS

$$I_{C(sat)} = V_{CC}/R_C \text{ (at } V_{CE} = 0) \qquad V_{CE(off)} = V_{CC} \text{ (at } I_C = 0) \qquad \text{slope} = -1/R_C$$

Wherever the load line crosses the curve for the actual I_B is the Q-point. For maximum symmetrical swing, place it near the middle ($V_{CEQ} \approx V_{CC}/2$).

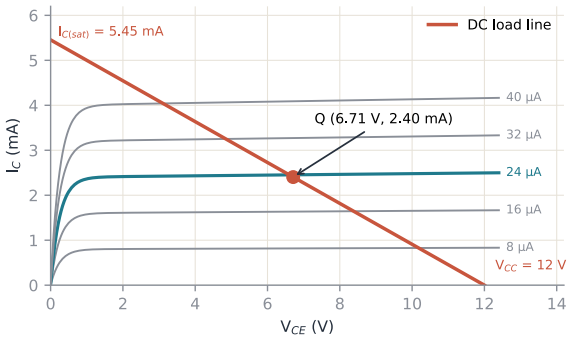


Fig 1.9 DC load line and Q-point for Example 1.5 (V_{CC} = 12 V, R_C = 2.2 kΩ, I_B = 24 µA).

Why the Q-point drifts — stability factor

The Q-point moves when temperature changes, because three things change: I_{CO} (leakage, doubles every 10 °C), V_{BE} (falls ≈ 2.5 mV/°C) and β (rises). Also two "identical" transistors can have β differing by 2–3×. Since $I_C = \beta I_B + (1 + \beta)I_{CO}$, we define

STABILITY FACTOR

$$S = \frac{\partial I_C}{\partial I_{CO}} = \frac{1+\beta}{1-\beta\,(\partial I_B/\partial I_C)} \qquad (\text{also } S' = \partial I_C/\partial V_{BE},\ S'' = \partial I_C/\partial \beta)$$

Smaller S = more stable. The best possible is S = 1. If heat raises I_C, which heats the junction more, which raises I_C further... that runaway loop is **thermal runaway**, and good bias breaks it.

Fixed bias (base bias)

A single resistor R_B from V_{CC} feeds the base (Fig 1.10a). KVL round the base loop:

FIXED BIAS

$$I_B = \frac{V_{CC} - V_{BE}}{R_B},\quad I_C = \beta I_B,\quad V_{CE} = V_{CC} - I_C R_C,\quad S = 1+\beta$$

I_B is fixed, so $\partial I_B/\partial I_C = 0$ and $S = 1 + \beta$ (≈ 101 — terrible). I_C is directly proportional to β. Simple, but rarely used in real designs.

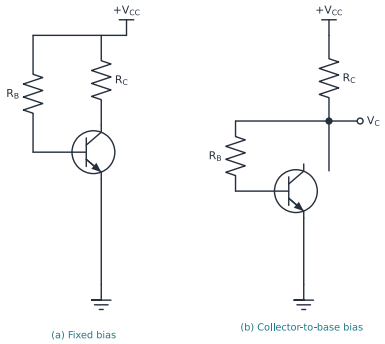


Fig 1.10 (a) Fixed bias and (b) collector-to-base (collector-feedback) bias.

SOLVED EXAMPLE 1.5 · FIXED BIAS

Fixed bias: V_{CC} = 12 V, R_B = 470 kΩ, R_C = 2.2 kΩ, β = 100, V_{BE} = 0.7 V. Find the Q-point and S. Repeat for β = 150.

- $I_B = (12 - 0.7)/470\text{ k} = 24.0\ \mu\text{A}$; $I_C = 100 \times 24.0\ \mu\text{A} = 2.40$ mA.
- $V_{CE} = 12 - 2.40\text{ m} \times 2.2\text{ k} = 6.71$ V. $S = 1 + \beta = 101$.
- β = 150: $I_C = 3.61$ mA (+50 %), $V_{CE} = 12 - 3.61 \times 2.2 = 4.07$ V. The Q-point moved a lot.

Q = (6.71 V, 2.40 mA), S = 101; with β = 150: (4.07 V, 3.61 mA)

Collector-to-base (collector-feedback) bias

Connect R_B from the **collector** instead of V_{CC} (Fig 1.10b). If I_C tries to rise, V_C drops (more drop on R_C), which reduces I_B, pulling I_C back — negative feedback.

COLLECTOR-TO-BASE BIAS

$$V_{CC} = (I_C + I_B)R_C + I_B R_B + V_{BE} \quad\Rightarrow\quad I_B = \frac{V_{CC} - V_{BE}}{R_B + (1+\beta)R_C}$$

$$\partial I_B/\partial I_C = -\frac{R_C}{R_C + R_B} \quad\Rightarrow\quad S = \frac{1+\beta}{1+\beta\ R_C/(R_C + R_B)}$$

Bonus: the transistor can never saturate hard, because $V_{CB} = I_B R_B > 0$ always.

SOLVED EXAMPLE 1.6 · COLLECTOR FEEDBACK

V_{CC} = 10 V, R_C = 4.7 kΩ, R_B = 250 kΩ, β = 90. Find I_C, V_{CE} and S. What is I_C if β rises to 135?

- $I_B = 9.3/(250\text{ k} + 91 \times 4.7\text{ k}) = 9.3/677.7\text{ k} = 13.7\ \mu\text{A} \rightarrow I_C = 90 \times 13.7\ \mu\text{A} = 1.235\ \text{mA}$.
- $V_{CE} = 10 - (1.235 + 0.0137)\text{ m} \times 4.7\text{ k} = 4.13\ \text{V}$.
- $S = 91/(1 + 90 \times 4.7/254.7) = 91/2.661 = 34.2$.
- β = 135: $I_B = 9.3/(250\text{ k} + 136 \times 4.7\text{ k}) = 10.46\ \mu\text{A} \rightarrow I_C = 1.41\ \text{mA}$ — only +14 % for a +50 % change in β.

I_C = 1.24 mA, V_{CE} = 4.13 V, S = 34.2; I_C(β = 135) = 1.41 mA

Voltage-divider bias — the one used in practice

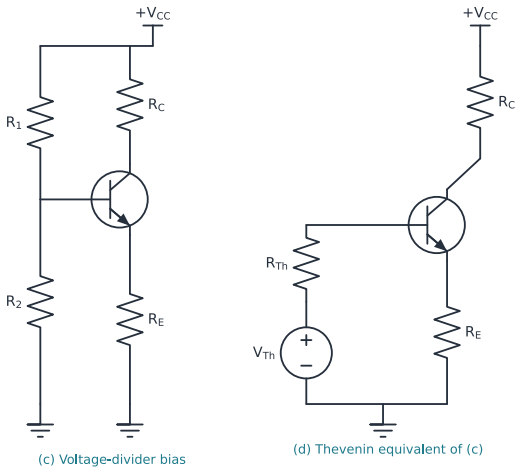


Fig 1.11 (c) Voltage-divider (self/emitter) bias and (d) its Thevenin equivalent at the base.

R_1–R_2 fix the base voltage; R_E provides negative feedback: if I_C rises, $V_E = I_E R_E$ rises, V_{BE} falls, I_B falls. Replace the divider by its Thevenin equivalent (Fig 1.11d):

VOLTAGE-DIVIDER BIAS

$$V_{Th} = V_{CC}\,\frac{R_2}{R_1 + R_2}, \quad R_{Th} = R_1 \parallel R_2$$

$$\text{Exact: } I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (1+\beta)R_E}, \quad V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$\text{Approximate (if } \beta R_E \geq 10 R_2\text{): } V_B \approx V_{Th}, \quad I_E \approx (V_B - V_{BE})/R_E, \quad V_{CE} \approx V_{CC} - I_C(R_C + R_E)$$

$$S = \frac{(1+\beta)(1 + R_{Th}/R_E)}{1 + \beta + R_{Th}/R_E} \quad \rightarrow 1 + R_{Th}/R_E \text{ when } \beta \text{ is large}$$

Derivation of S: from $V_{Th} = I_B R_{Th} + V_{BE} + (I_B + I_C)R_E$, differentiate w.r.t. I_C: $\partial I_B/\partial I_C = -R_E/(R_{Th} + R_E)$. Put it in the general S formula. The smaller R_{Th}/R_E, the closer S is to 1.

SOLVED EXAMPLE 1.7 · VOLTAGE-DIVIDER BIAS

V_{CC} = 22 V, R_1 = 39 kΩ, R_2 = 3.9 kΩ, R_C = 10 kΩ, R_E = 1.5 kΩ, β = 100. Find the Q-point exactly and approximately, and S.

- $V_{Th} = 22 \times 3.9/42.9 = 2.0$ V; $R_{Th} = 39\text{ k} \parallel 3.9\text{ k} = 3.55$ kΩ.
- Exact: $I_B = 1.3/(3.55\text{ k} + 101 \times 1.5\text{ k}) = 8.38$ μA → $I_C = 0.838$ mA, $V_{CE} = 22 - 0.838 \times 10 - 0.847 \times 1.5 = 12.35$ V.
- Check approximation: $\beta R_E = 150\text{ k} \geq 10R_2 = 39\text{ k}$ √. $V_E = 1.3$ V, $I_E = 0.867$ mA, $V_{CE} = 22 - 0.867 \times 11.5 = 12.03$ V.
- $S = 101 \times (1 + 2.364)/(101 + 2.364) = 3.29$ — compare 101 for fixed bias!

Exact (12.35 V, 0.838 mA); approx (12.03 V, 0.867 mA); S = 3.29

Bias circuit	Stability factor S	Comment
Fixed bias	$1 + \beta$ (≈ 101)	$I_C \propto \beta$; thermal runaway risk
Collector-to-base	$(1 + \beta)/(1 + \beta R_C/(R_C + R_B))$	Better; feedback also reduces AC gain
Voltage divider	$\approx 1 + R_{Th}/R_E$	Best; almost β-independent; used everywhere

INTERVIEW CORNER

Q. Why is voltage-divider bias called "β-independent"?
A. When $\beta R_E \gg R_{Th}$, $I_E \approx (V_{Th} - V_{BE})/R_E$, which contains no β at all.

Q. What is thermal runaway and how do you prevent it?
A. Positive thermal feedback: $I_C \uparrow$ → power ↑ → temperature ↑ → I_{CO}, β ↑ → $I_C \uparrow$. Prevent with emitter resistor/feedback bias (low S), heat sinks, and keeping $V_{CE} \leq V_{CC}/2$.

Q. Why not just make R_E huge for perfect stability?
A. It eats headroom (less V_{CE}, less swing) and kills AC gain unless bypassed with C_E.

1.8 MOSFET biasing

A MOSFET's gate draws **no DC current** ($I_G = 0$). That makes the gate side easy — the gate voltage is set by plain voltage division — but the drain current depends on V_{GS} through a **square law**, so we solve a quadratic and keep the root that makes physical sense.

DEVICE EQUATIONS (SATURATION)

Enhancement NMOS: $I_D = K(V_{GS} - V_T)^2$, valid for $V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$ ($K = \tfrac{1}{2}\mu_n C_{ox} W/L$); here V_T is the threshold voltage, not the 25 mV thermal voltage

Depletion NMOS / JFET: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$

Voltage-divider bias (Fig 1.12a)

$V_G = V_{DD}R_2/(R_1 + R_2)$ (no gate current, so this is exact). The source resistor gives $V_{GS} = V_G - I_D R_S$. Substitute into the square law and solve. R_S gives the same negative-feedback stabilisation as R_E did for the BJT. R_1, R_2 can be megohms, so input resistance stays high.

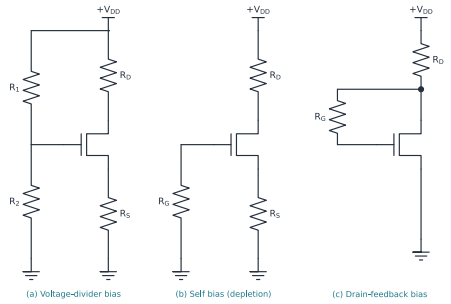


Fig 1.12 MOSFET bias circuits.

Self bias (Fig 1.12b)

The gate is tied to ground through R_G (no current, so $V_G = 0$). Drain current through R_S lifts the source, making $V_{GS} = -I_D R_S$ — **negative**. This works only for **depletion-mode** MOSFETs (and JFETs), which conduct at negative V_{GS}. An enhancement MOSFET needs $V_{GS} > V_T > 0$, so it would simply stay off.

Drain-feedback bias (Fig 1.12c)

R_G connects drain to gate. No gate current means no drop across R_G, so $\mathbf{V_G = V_D}$, i.e. $V_{GS} = V_{DS} = V_{DD} - I_D R_D$. Since $V_{DS} = V_{GS} > V_{GS} - V_T$, the transistor is **automatically in saturation**. If I_D rises, V_D falls, V_{GS} falls → I_D pulled back.

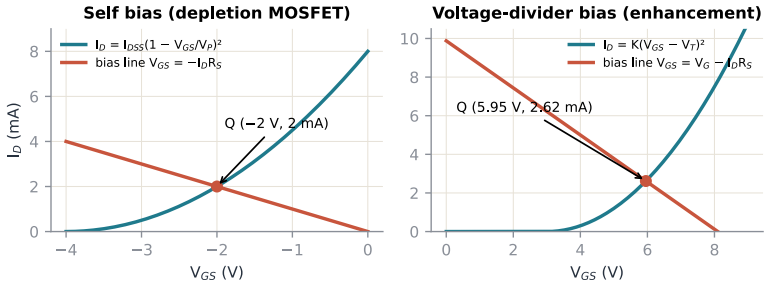


Fig 1.13 Graphical Q-point: the bias line meets the transfer curve (Examples 1.9 and 1.8).

SOLVED EXAMPLE 1.8 · MOSFET VOLTAGE-DIVIDER BIAS

Enhancement NMOS: $V_T = 3$ V, $K = 0.3$ mA/V². $V_{DD} = 18$ V, $R_1 = 22$ MΩ, $R_2 = 18$ MΩ, $R_D = 3$ kΩ, $R_S = 0.82$ kΩ. Find I_{DQ}, V_{GSQ}, V_{DSQ}.

- $V_G = 18 \times 18/40 = 8.1$ V; $V_{GS} = 8.1 - 0.82I_D$ (I_D in mA).
- $I_D = 0.3(5.1 - 0.82I_D)^2 \rightarrow 0.2017I_D^2 - 3.509I_D + 7.803 = 0$.

- Roots: 14.78 mA or 2.62 mA. 14.78 mA gives $V_{GS} = -4.0$ V < V_T (device off) — reject.
- $I_D = 2.62$ mA → $V_{GS} = 8.1 - 2.15 = 5.95$ V; $V_{DS} = 18 - 2.62 \times 3.82 = 8.0$ V.
- Saturation check: $V_{DS} = 8.0 \geq V_{GS} - V_T = 2.95$ √.

$I_{DQ} = 2.62$ mA, $V_{GSQ} = 5.95$ V, $V_{DSQ} = 8.0$ V

SOLVED EXAMPLE 1.9 · SELF BIAS (DEPLETION MOSFET)

Depletion NMOS: $I_{DSS} = 8$ mA, $V_P = -4$ V. $R_S = 1$ kΩ, $R_D = 1.2$ kΩ, $V_{DD} = 16$ V. Find the Q-point.

- $V_{GS} = -I_D R_S = -I_D$ (mA, V). $I_D = 8(1 - I_D/4)^2$ → $0.5I_D^2 - 5I_D + 8 = 0$.
- Roots: 8 mA or 2 mA. 8 mA would need $V_{GS} = -8$ V, beyond pinch-off — reject.
- $I_D = 2$ mA, $V_{GS} = -2$ V, $V_{DS} = 16 - 2 \times 2.2 = 11.6$ V.

Q = ($V_{GS} = -2$ V, $I_D = 2$ mA), $V_{DS} = 11.6$ V

SOLVED EXAMPLE 1.10 · DRAIN-FEEDBACK BIAS

Enhancement NMOS with $K = 0.5$ mA/V², $V_T = 2$ V in Fig 1.12c; $V_{DD} = 12$ V, $R_D = 2$ kΩ, $R_G = 10$ MΩ. Find I_D and V_{DS}.

- $V_{GS} = V_{DS} = 12 - 2I_D$. $I_D = 0.5(10 - 2I_D)^2$ → $2I_D^2 - 21I_D + 50 = 0$.
- Roots: 6.85 mA (gives $V_{GS} = -1.7$ V, reject) or 3.65 mA.
- $I_D = 3.65$ mA → $V_{GS} = V_{DS} = 12 - 7.30 = 4.70$ V (in saturation automatically).

$I_D = 3.65$ mA, $V_{DS} = V_{GS} = 4.70$ V

COMMON MISTAKE

Keeping the wrong root of the quadratic. Always test each root: for enhancement V_{GS} must exceed V_T; for depletion V_{GS} must lie between V_P and 0 (or slightly positive).

INTERVIEW CORNER

Q. Why can't self bias be used with an enhancement MOSFET?

A. Self bias makes $V_{GS} = -I_D R_S \leq 0$, but an enhancement NMOS needs $V_{GS} > V_T > 0$ to conduct.

Q. A diode-connected MOSFET (gate tied to drain) — which region is it in?

A. Always saturation (if on), because $V_{DS} = V_{GS} > V_{GS} - V_T$. Drain-feedback bias exploits the same fact.

✓ Module 1 — Quick revision

Time constant: $\tau = RC$; 63 % at 1τ, ~100 % at 5τ.

HPF step: $V\,e^{-t/RC}$. **LPF step:** $V(1 - e^{-t/RC})$.

Differentiator: output across R, $RC \ll T$, $v_o \approx RC\,dv_i/dt$.

Integrator: output across C, $RC \gg T$, $v_o \approx (1/RC)\int v_i dt$.

Square into HPF: $V_1 = V/(1 + e^{-T/2RC})$, $V_1' = V_1 e^{-T/2RC}$; tilt $\approx T/2RC = \pi f_1/f$.

Pulse tilt: $\approx t_p/RC$. **Rise time:** $t_r = 2.2RC = 0.35/f_2$.

Filters: $f_c = 1/2\pi RC$; −3 dB, ±45° at f_c; 20 dB/decade.

Clipper: changes shape. Shunt biased: clips at $V_R + V_\gamma$. Double-ended: two levels.

Clamper: changes DC level only; needs $R_L C \gg T/2$; peak-to-peak unchanged.

Load line: $V_{CC} = I_C R_C + V_{CE}$; ends V_{CC}/R_C and V_{CC}.

$S = \partial I_C/\partial I_{CO}$: fixed $1 + \beta$; collector-base $(1 + \beta)/(1 + \beta R_C/(R_C + R_B))$; VDB $\approx 1 + R_{Th}/R_E$.

VDB approximation valid when $\beta R_E \geq 10 R_2$.

MOSFET: $I_D = K(V_{GS} - V_T)^2$; depletion $I_{DSS}(1 - V_{GS}/V_P)^2$.

Self bias $V_{GS} = -I_D R_S$ (depletion only). **Drain feedback** $V_{GS} = V_{DS}$ (always saturated).

Practice — short questions (3 marks)

1. Draw the output of an RC high-pass circuit for a square-wave input when $RC \ll T$, $RC = T$ and $RC \gg T$.
2. Show that an RC low-pass circuit acts as an integrator. State the condition.
3. Define tilt. A 2 kHz square wave is applied to a high-pass circuit with RC = 5 ms. Find the tilt. [≈ 5 %]
4. Derive the rise-time–bandwidth relation of a low-pass RC circuit.
5. Differentiate between clippers and clampers with one circuit each.
6. Draw the transfer characteristic of a shunt clipper biased at +5 V (silicon diode). [clips at 5.7 V]
7. Why is self bias suitable only for depletion MOSFETs?

Practice — long questions (9/14 marks)

1. Derive the steady-state output levels of an RC high-pass circuit driven by a symmetric square wave. Hence obtain the percentage tilt. A 500 Hz, 20 V_{pp} square wave drives R = 4.7 kΩ, C = 1 µF; find V_1, V_1'. [≈ 11.06 V, 8.94 V]
2. Design a double-ended clipper that limits a 12 V peak sine wave between −3 V and +5 V using silicon diodes. Draw the transfer characteristic and the output. [V_1 = 4.3 V, V_2 = 2.3 V]
3. Explain the working of a biased negative clamper with waveforms. What happens if the time constant is too small?
4. Derive the expression for stability factor S of a voltage-divider bias circuit. Compute S, I_{CQ}, V_{CEQ} for V_{CC} = 15 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 2.2 kΩ, R_E = 1 kΩ, β = 120. [S ≈ 8.7, I_C ≈ 1.79 mA, V_{CE} ≈ 9.25 V]
5. For a drain-feedback biased NMOS (K = 0.25 mA/V², V_T = 1.5 V, V_{DD} = 10 V, R_D = 4.7 kΩ) find I_D and V_{DS}. Explain why the device is always saturated. [I_D ≈ 1.32 mA, V_{DS} ≈ 3.80 V]

02

BJT & MOSFET Amplifiers

> *Gain, input/output resistance, bandwidth, Miller effect and the cascode — this is the vocabulary of every analog and VLSI interview, and GATE asks a numerical on it every year.*

SYLLABUS CHECKLIST

- ☐ RC-coupled CE amplifier
- ☐ Small-signal analysis using the hybrid-π model — gain, R_{in}, R_{out}
- ☐ Frequency response — low, mid and high frequency
- ☐ Miller effect and Miller capacitance
- ☐ Short-circuit current gain, f_β and f_T
- ☐ MOSFET small-signal model; CS amplifier with resistive, current-source and diode-connected loads
- ☐ Multistage amplifiers — cascade, gain and bandwidth
- ☐ Cascode amplifier (BJT/MOS) and its advantages

Module 2 · CO2 · Analyse small-signal gain and frequency response of BJT and MOSFET amplifiers

2.1 The RC-coupled CE amplifier

An amplifier takes a small wiggle at its input and produces a **bigger copy** at its output. The transistor does not create energy — it acts like a valve, letting the DC supply push a large current whose shape follows the tiny input signal.

> **THINK OF IT LIKE**
>
> A tap: a small twist of your wrist (input) controls a big flow of water (output). The water comes from the mains (V_{CC}), not from your wrist.

Every part of Fig 2.1 has one job:

Component	Job
R_1, R_2, R_E	Voltage-divider bias — sets a stable Q-point (Module 1)
R_C	Converts the collector current swing into a voltage swing
C_{C1}, C_{C2} (coupling)	Pass AC, block DC, so the source and load don't disturb the bias
C_E (bypass)	Shorts R_E for AC: keeps DC stability without losing AC gain

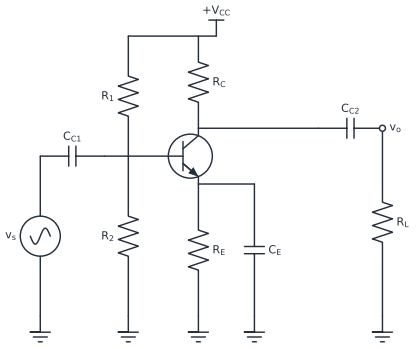


Fig 2.1 Single-stage RC-coupled common-emitter amplifier.

The output is **180° out of phase** with the input: when the base voltage rises, I_C rises, the drop on R_C rises, so V_C falls. Throughout this module we use one design as a running example: **V_{CC} = 12 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 4.7 kΩ, R_E = 1.3 kΩ, R_L = 10 kΩ, R_s = 1 kΩ, β = 100, V_A = 100 V**. Its DC analysis (exact method of Module 1) gives I_C = 1.01 mA ≈ 1 mA and V_{CE} = 5.94 V.

2.2 Hybrid-π small-signal model

For *small* signals around the Q-point, a curved transistor characteristic looks like a straight line, so we can replace the transistor with resistors and a controlled source. The **hybrid-π model** (Fig 2.2) is the most physical version.

HYBRID-Π PARAMETERS

$$g_m = \frac{I_C}{V_T} \quad r_\pi = \frac{\beta}{g_m} \quad r_o = \frac{V_A}{I_C} \quad r_e = \frac{V_T}{I_E} \approx \frac{1}{g_m}$$

- **g_m** (transconductance) — how many mA of collector current per volt of v_{be}. At 1 mA it is 40 mA/V. It depends only on I_C, not on the transistor type!
- **r_π** — resistance looking into the base (emitter grounded). **r_x** — base-spreading resistance (tens of Ω).
- **r_o** — models the Early effect (slight upward slope of the output curves).
- **C_π** — base-emitter diffusion + junction capacitance (≈ 5–20 pF). **C_μ** — collector-base junction capacitance (≈ 1–3 pF), small but dangerous (Miller!).

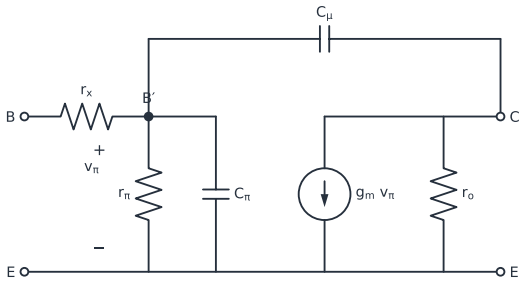


Fig 2.2 High-frequency hybrid-π model. At mid-band drop C_π, C_μ (open) and usually r_x.

Link with h-parameters: $h_{ie} \approx r_x + r_\pi$, $h_{fe} = g_m r_\pi = \beta$, $h_{oe} \approx 1/r_o$.

2.3 Mid-band analysis: gain, R_{in}, R_{out}

In the **mid band** the coupling and bypass capacitors are short circuits and the tiny internal capacitances are open circuits. The DC supply is an AC ground. That leaves Fig 2.3, where $R_B = R_1 \parallel R_2$.

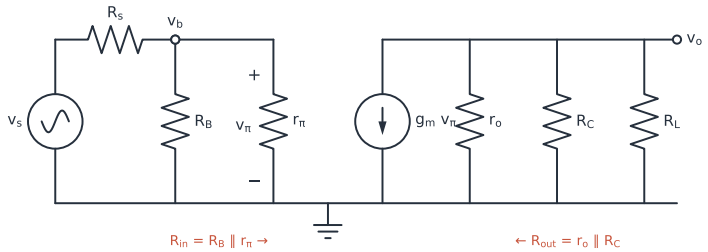


Fig 2.3 Mid-band small-signal equivalent of the CE amplifier (C_E bypasses R_E).

CE AMPLIFIER, MID-BAND

$$A_v = \frac{v_o}{v_b} = -g_m(r_o \parallel R_C \parallel R_L) \quad R_{in} = R_B \parallel r_\pi \quad R_{out} = r_o \parallel R_C$$

$$\text{Overall: } A_{vs} = \frac{v_o}{v_s} = A_v \cdot \frac{R_{in}}{R_{in} + R_s}$$

$$\text{With } R_E \text{ un-bypassed: } A_v \approx -\frac{R_C \parallel R_L}{r_e + R_E}, \quad R_{in} = R_B \parallel (\beta + 1)(r_e + R_E)$$

Derivation in one line: $v_\pi = v_b$; the current $g_m v_\pi$ flows (downwards) out of the parallel combination $r_o \parallel R_C \parallel R_L$, so $v_o = -g_m v_b (r_o \parallel R_C \parallel R_L)$.

SOLVED EXAMPLE 2.1 · MID-BAND GAIN

For the running example (I_C = 1 mA, β = 100, V_A = 100 V, C_E bypassing R_E) find g_m, r_π, r_o, R_{in}, R_{out}, A_v and A_{vs}.

- g_m = 1 mA/25 mV = 40 mS; r_π = 100/0.04 = 2.5 kΩ; r_o = 100/1 mA = 100 kΩ.
- R_B = 47 k ∥ 10 k = 8.25 kΩ → R_{in} = 8.25 k ∥ 2.5 k = 1.92 kΩ.
- R_L' = 100 k ∥ 4.7 k ∥ 10 k = 3.10 kΩ → A_v = −0.04 × 3098 = −123.9.
- A_{vs} = −123.9 × 1.92/(1.92 + 1) = −81.5 (38.2 dB). R_{out} = 100 k ∥ 4.7 k = 4.49 kΩ.

A_v = −124, A_{vs} = −81.5, R_{in} = 1.92 kΩ, R_{out} = 4.49 kΩ

2.4 Frequency response — low, mid and high

Plot the gain against frequency (log scale) and you get the familiar "flat table with sloping ends" (Fig 2.4):

- **Low band:** the coupling and bypass capacitors are no longer short circuits. Each one forms a high-pass RC with the resistance it "sees", so gain falls as frequency falls.
- **Mid band:** all capacitors ignored — gain is flat (Section 2.3).
- **High band:** the internal C_π and C_μ start shorting the signal, so gain falls as frequency rises.

Bandwidth BW = $f_H - f_L \approx f_H$; the product $|A_{mid}| \times$ BW is the **gain-bandwidth product**.

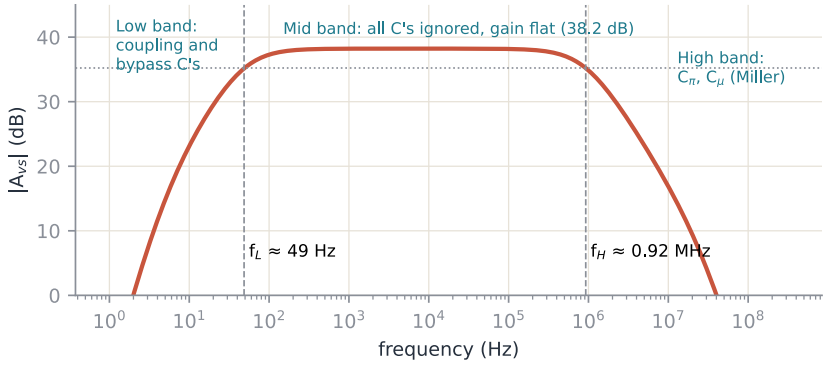


Fig 2.4 Frequency response of the running example, computed from the poles found in Examples 2.2 and 2.3.

Low-frequency cut-offs

Treat one capacitor at a time (others shorted) and find the resistance it sees:

LOW-FREQUENCY POLES

$$f_{L1} = \frac{1}{2\pi(R_s + R_{in})C_{C1}} \qquad f_{L2} = \frac{1}{2\pi(R_{out} + R_L)C_{C2}} \qquad f_{LE} = \frac{1}{2\pi R_e' C_E}$$

$$R_e' = R_E \parallel \frac{r_\pi + (R_s \parallel R_B)}{\beta + 1} \qquad f_L \approx \text{largest of these (or their sum, as an estimate)}$$

The resistance seen by C_E is tiny (looking into the emitter divides everything by $\beta + 1$), so **C_E usually sets f_L**. That is why bypass capacitors are big (100 µF or more).

SOLVED EXAMPLE 2.2 · LOW-FREQUENCY RESPONSE

In the running example $C_{C1} = C_{C2} = 10$ µF and $C_E = 100$ µF. Find f_{L1}, f_{L2}, f_{LE} and estimate f_L.

- $f_{L1} = 1/(2\pi \times 2.92\text{ k} \times 10\ \mu) = 5.45$ Hz.
- $f_{L2} = 1/(2\pi \times (4.49\text{ k} + 10\text{ k}) \times 10\ \mu) = 1.10$ Hz.
- $R_s \parallel R_B = 0.892$ kΩ; $(2.5\text{ k} + 0.892\text{ k})/101 = 33.6\ \Omega$; $R_e' = 1.3\text{ k} \parallel 33.6 = 32.7\ \Omega \rightarrow f_{LE} = 1/(2\pi \times 32.7 \times 100\ \mu) = 48.6$ Hz.
- C_E dominates: $f_L \approx 49$ Hz (an exact solve gives 49.2 Hz; the sum-estimate gives 55 Hz).

$f_{L1} = 5.45$ Hz, $f_{L2} = 1.10$ Hz, $f_{LE} = 48.6$ Hz → $f_L \approx 49$ Hz

2.5 Miller effect and Miller capacitance

Here is the surprise of high-frequency design: the tiny C_μ (2 pF) between base and collector behaves like a **huge** capacitor at the input. Why? The collector swings A times more than the base, and in the opposite direction. So the voltage across C_μ is $v_b(1 + A)$ — the capacitor must be charged $(1 + A)$ times harder.

THINK OF IT LIKE

Pushing a door shut while someone on the other side pulls it open 100 times harder. It feels like a door 100 times heavier.

Derivation. Current into C from the input side: $I = j\omega C(V_1 - V_2) = j\omega C\, V_1(1 + A)$. So, seen from the input, it is a capacitor $C(1 + A)$ to ground. From the output side: $I = j\omega C(V_2 - V_1) = j\omega C\, V_2(1 + 1/A)$, a capacitor $C(1 + 1/A) \approx C$.

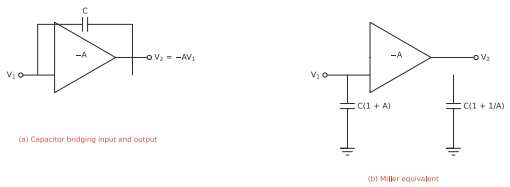


Fig 2.5 Miller's theorem: a bridging capacitor across an inverting gain −A splits into two grounded capacitors.

MILLER CAPACITANCE (CE STAGE)

$$C_M = C_\mu(1 + g_m R_L') \quad C_{in} = C_\pi + C_M \quad C_{out} = C_\mu(1 + 1/g_m R_L') \approx C_\mu$$

$$f_H \approx \frac{1}{2\pi\, R_{sig}'\, C_{in}}, \quad R_{sig}' = r_\pi \parallel (r_x + R_s \parallel R_B)$$ (output pole $f_{H2} = 1/2\pi R_L' C_{out}$ is usually much higher)

SOLVED EXAMPLE 2.3 · HIGH-FREQUENCY RESPONSE

For the running example C_π = 12 pF, C_μ = 2 pF, $r_x \approx 0$. Find the Miller capacitance, C_{in}, f_H and the gain-bandwidth product.

- $g_m R_L'$ = 0.04 × 3098 = 123.9 → C_M = 2 p × 124.9 = 249.9 pF (125× bigger than C_μ!).
- C_{in} = 12 + 249.9 = 261.9 pF. R_{sig}' = 1 k ∥ 8.25 k ∥ 2.5 k = 657 Ω.
- f_H = 1/(2π × 657 × 261.9 p) = 925 kHz. Output pole: C_{out} = 2.02 pF, f_{H2} = 1/(2π × 3098 × 2.02 p) = 25.5 MHz — not dominant.
- GBW = 81.5 × 0.925 MHz ≈ 75 MHz.

$C_M \approx 250$ pF, $f_H \approx 0.92$ MHz, GBW ≈ 75 MHz

2.6 Short-circuit current gain, f_β and f_T

How fast is the transistor *itself*, independent of circuit resistors? Short the output (collector to emitter) and measure i_c/i_b. With the output shorted there is no Miller multiplication, and from Fig 2.2:

SHORT-CIRCUIT CURRENT GAIN

$$h_{fe}(j\omega) = \frac{i_c}{i_b} \approx \frac{g_m r_\pi}{1 + j\omega r_\pi (C_\pi + C_\mu)} = \frac{\beta}{1 + j\, f/f_\beta}$$

$$f_\beta = \frac{1}{2\pi\, r_\pi (C_\pi + C_\mu)} \qquad f_T = \beta f_\beta = \frac{g_m}{2\pi (C_\pi + C_\mu)}$$

Derivation: i_b flows into $r_\pi \parallel (C_\pi + C_\mu)$ (C_μ is grounded at the collector side because the output is shorted), so $v_\pi = i_b\, r_\pi/(1 + j\omega r_\pi(C_\pi + C_\mu))$ and $i_c \approx g_m v_\pi$ (neglecting the small current through C_μ).

- **f_β** (beta cut-off): where $|h_{fe}|$ falls to $\beta/\sqrt{2}$.
- **f_T** (unity-gain or transition frequency): where $|h_{fe}| = 1$. Beyond f_β the gain falls at 20 dB/decade, so $|h_{fe}| \times f = f_T$ — the transistor's gain-bandwidth product.

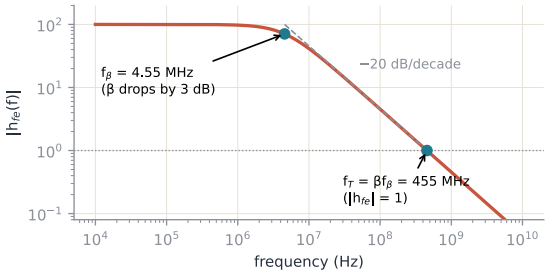


Fig 2.6 $|h_{fe}|$ vs frequency for Example 2.4.

SOLVED EXAMPLE 2.4 · F_T AND F_B

A BJT at I_C = 1 mA has β = 100, C_π = 12 pF, C_μ = 2 pF. Find f_T, f_β, and $|h_{fe}|$ at 50 MHz.

- $f_T = 0.04/(2\pi \times 14\text{ p}) = 455$ MHz; $f_\beta = 455/100 = 4.55$ MHz (same as $1/(2\pi \times 2.5\text{ k} \times 14\text{ p})$).
- At 50 MHz ($\gg f_\beta$): $|h_{fe}| \approx f_T/f = 455/50 \approx 9.1$.

f_T = 455 MHz, f_β = 4.55 MHz, $|h_{fe}|$(50 MHz) ≈ 9.1

INTERVIEW CORNER

Q. Why does f_T increase with I_C (up to a point)?

A. $g_m \propto I_C$ while capacitances grow more slowly, so $f_T = g_m/2\pi(C_\pi + C_\mu)$ rises; at very high current, high-injection effects make it fall again.

Q. What limits the bandwidth of a CE amplifier most?

A. The Miller-multiplied C_μ at the input combined with the source resistance — the dominant high-frequency pole.

Q. Why is CB or CC amplifier bandwidth higher than CE?

A. Neither has an inverting gain across C_μ at its input, so there is no Miller multiplication.

2.7 MOSFET small-signal model

The MOSFET model is even simpler than the BJT's: the gate is insulated, so the gate terminal is an **open circuit** — no r_π. A voltage-controlled current source $g_m v_{gs}$ and r_o are all that remain (Fig 2.7; at high frequency add C_{gs} and C_{gd}, the MOS versions of C_π and C_μ).

MOSFET PARAMETERS ($I_D = K(V_{GS} - V_T)^2$)

$$g_m = 2K(V_{GS} - V_T) = \frac{2I_D}{V_{GS} - V_T} = 2\sqrt{(K\, I_D)} \qquad r_o = \frac{1}{\lambda I_D}$$

intrinsic gain $A_0 = g_m r_o$

KEY IDEA

For the same current a BJT gives more g_m ($I_C/0.025$ V) than a MOSFET ($2I_D/V_{ov}$, with $V_{ov} = V_{GS} - V_T \approx 0.2$–$0.5$ V). The MOSFET wins on input resistance (infinite) and on scaling in ICs.

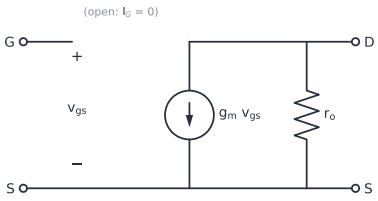


Fig 2.7 Low-frequency MOSFET small-signal model.

2.8 CS amplifier: resistive, current-source and diode-connected loads

The common-source (CS) stage is the MOS twin of the CE stage: input at gate, output at drain, source at AC ground. Its gain is always "$-g_m \times$ (total resistance at the drain)". What changes is the **load** (Fig 2.8).

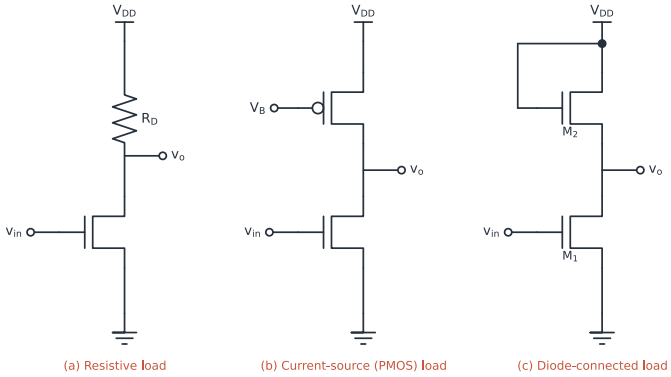


Fig 2.8 Common-source amplifier with three kinds of load.

(a) Resistive load

$A_v = -g_m(R_D \parallel r_o)$, $R_{in} = \infty$ (or the bias resistors), $R_{out} = R_D \parallel r_o$. Raising R_D raises gain, but also raises the DC drop $I_D R_D$ — you run out of supply headroom. In ICs, big resistors also waste area.

(b) Current-source (active) load

Replace R_D by a PMOS biased in saturation by a fixed V_B. It supplies a constant DC current (only a small V_{SD} drop needed) but presents its large r_{o2} to the signal:

$$A_v = -g_{m1}(r_{o1} \parallel r_{o2}) \quad (\approx -g_m r_o/2 \text{ for equal } r_o)$$

This is how real IC amplifiers get gains of 50–100 from a single stage.

(c) Diode-connected load

M_2 has gate tied to drain, so it is always saturated, and its small-signal resistance is just $\mathbf{1/g_{m2}}$ (a current $g_{m2}v$ flows when voltage v is applied). Ignoring r_o and body effect:

$$A_v = -\frac{g_{m1}}{g_{m2}} = -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \quad \text{(same } I_D\text{, both NMOS)}$$

The gain is low but set by a **ratio of transistor sizes** — very accurate and nearly independent of process, temperature and supply. Used where precise moderate gain is needed.

SOLVED EXAMPLE 2.5 · THREE CS LOADS

An NMOS with K = 1 mA/V², λ = 0.02 V^{-1} is biased at I_D = 0.5 mA. Find the gain with (a) R_D = 10 kΩ, (b) a PMOS current-source load with λ_p = 0.02 V^{-1}, (c) a diode-connected NMOS load with $(W/L)_1/(W/L)_2$ = 25.

- $V_{ov} = \sqrt{(0.5/1)} = 0.707$ V; $g_m = 2\sqrt{(1 \times 0.5)} = 1.414$ mS; $r_o = 1/(0.02 \times 0.5\text{ m}) = 100$ kΩ.
- (a) $A_v = -1.414\text{ m} \times (10\text{ k} \parallel 100\text{ k}) = -1.414 \times 9.09 = -12.9$.
- (b) $r_{o2} = 100$ kΩ → $A_v = -1.414\text{ m} \times 50\text{ k} = -70.7$ (half the intrinsic gain $g_m r_o = 141$).
- (c) $A_v = -\sqrt{25} = -5$.

(a) −12.9 (b) −70.7 (c) −5

2.9 Multistage (cascade) amplifiers

One stage rarely gives enough gain, or the right input and output resistance. So we chain stages: the output of one drives the input of the next. In RC coupling, a capacitor joins the stages so each keeps its own bias.

CASCADE

$$A_v = A_1 \times A_2 \times \ldots \times A_n \qquad A_v(\text{dB}) = A_1(\text{dB}) + A_2(\text{dB}) + \ldots \qquad R_{in} = R_{in1},\ R_{out} = R_{out,n}$$

$$\text{n identical stages: } f_H(n) = f_H\sqrt{(2^{1/n} - 1)} \qquad f_L(n) = \frac{f_L}{\sqrt{(2^{1/n} - 1)}}$$

Derivation of bandwidth shrinkage. Each stage has $|A/A_m| = 1/\sqrt{(1 + (f/f_H)^2)}$. For n stages the product is $[1 + (f/f_H)^2]^{-n/2}$. Set this equal to $1/\sqrt{2}$: $[1 + (f/f_H)^2]^n = 2$ → $f = f_H\sqrt{(2^{1/n} - 1)}$. The same steps on the low side give $f_L(n)$.

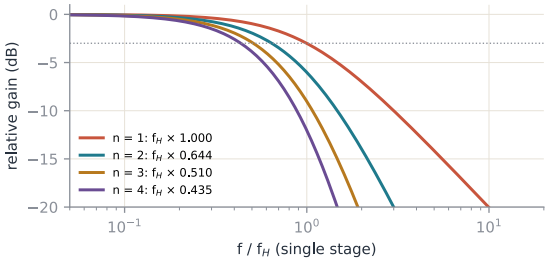


Fig 2.9 Adding identical stages multiplies gain but narrows the bandwidth.

KEY IDEA

Remember the loading effect: the gain of stage k must be calculated **with stage k + 1's input resistance as its load**. Multiplying "unloaded" gains overestimates the total.

SOLVED EXAMPLE 2.6 · THREE-STAGE CASCADE

Three identical stages each have mid-band gain −20, f_L = 50 Hz and f_H = 1 MHz (loading already included). Find the overall gain (ratio and dB), f_L, f_H.

- $A_v = (-20)^3 = -8000 \rightarrow 20 \log 8000 = 78.1$ dB (= 3 × 26.02 dB √).
- $\sqrt{(2^{1/3} - 1)} = \sqrt{0.2599} = 0.510$.
- $f_H = 1\text{ MHz} \times 0.510 = 510$ kHz; $f_L = 50/0.510 = 98.1$ Hz.

−8000 (78.1 dB), f_L = 98 Hz, f_H = 510 kHz

2.10 Cascode amplifier

A cascode stacks a **common-base (or common-gate) transistor on top of a common-emitter (or common-source) transistor** (Fig 2.10). The lower device converts input voltage to current; the upper device simply passes that current to the output while holding the lower device's collector (drain) at an almost fixed voltage.

Why it is so popular

1. **Kills the Miller effect.** Q_1's load is the emitter of Q_2, whose resistance is only $1/g_{m2}$. So Q_1's gain is $-g_{m1}/g_{m2} \approx -1$ and its $C_{\mu 1}$ is multiplied only by 2. Bandwidth goes up enormously.
2. **Same overall gain as CE.** The full signal current still reaches R_C: $A_v \approx -g_{m1}(R_C \parallel R_L)$.
3. **Very high output resistance.** $R_{out} \approx \beta\, r_o$ (BJT) or $\approx g_{m2} r_{o2} r_{o1}$ (MOS) — so with a current-source load the gain approaches $(g_m r_o)^2$.
4. **Good isolation** (reverse transmission is tiny) — useful in RF amplifiers and mixers.

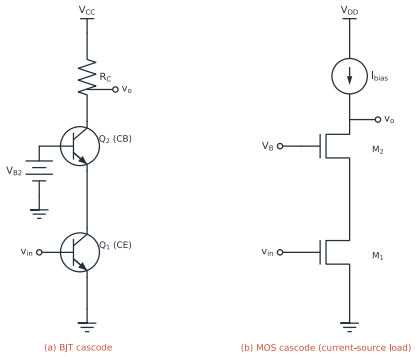


Fig 2.10 BJT and MOS cascode amplifiers (bias details simplified).

The cost: one extra transistor and extra headroom (one more $V_{CE(sat)}$ or V_{ov} in the stack).

MOS CASCODE OUTPUT RESISTANCE

$$R_{out} = r_{o1} + r_{o2} + g_{m2} r_{o2} r_{o1} \approx g_{m2} r_{o2} r_{o1} \qquad A_v = -g_{m1} R_{out} \text{ (ideal current-source load)}$$

SOLVED EXAMPLE 2.7 · CASCODE BENEFITS

(a) For the BJT of Example 2.3 (C_μ = 2 pF, CE gain −124), compare the Miller capacitance of a plain CE stage with that of a cascode. (b) A MOS cascode uses two transistors each with g_m = 1.414 mS and r_o = 100 kΩ, with an ideal current-source load. Find R_{out} and the gain.

- (a) CE: $C_M = 2\text{ p} \times 124.9 \approx 250$ pF. Cascode: Q_1 gain ≈ −1 → $C_M = 2\text{ p} \times 2 = 4$ pF — over 60× smaller.
- (b) $R_{out} = 100\text{ k} + 100\text{ k} + 1.414\text{ m} \times 100\text{ k} \times 100\text{ k} = 14.3$ MΩ.
- $A_v = -1.414\text{ m} \times 14.34\text{ M} \approx -20\,300 \approx -(g_m r_o)^2 = -141.4^2 = -20\,000$.

(a) 250 pF → 4 pF (b) R_{out} ≈ 14.3 MΩ, A_v ≈ −2 × 10⁴

INTERVIEW CORNER

Q. Why is a diode-connected MOSFET always in saturation, and what is its small-signal resistance?

A. $V_{DS} = V_{GS} > V_{GS} - V_T$. Its resistance is $1/g_m$ ($\| r_o$).

Q. What is the intrinsic gain of a MOSFET and how do you increase it?

A. $g_m r_o = 2/(\lambda V_{ov})$. Use longer channel (smaller λ), lower V_{ov}, or cascoding.

Q. Why don't we use big drain resistors in ICs for high gain?

A. They take large area and huge DC drop (headroom). An active (current-source) load gives high AC resistance with small DC drop.

✓ Module 2 — Quick revision

Hybrid-π: $g_m = I_C/V_T$ (40 mS at 1 mA); $r_\pi = \beta/g_m$; $r_o = V_A/I_C$.

CE mid-band: $A_v = -g_m(r_o \parallel R_C \parallel R_L)$; $R_{in} = R_B \parallel r_\pi$; $R_{out} = r_o \parallel R_C$.

Unbypassed R_E: $A_v \approx -R_C'/(r_e + R_E)$; R_{in} ↑ by $(\beta + 1)R_E$.

Low-f poles: $1/2\pi R_{seen}C$; C_E sees $R_E \parallel (r_\pi + R_s \| R_B)/(\beta + 1)$ → usually dominant.

Miller: $C_{in} = C(1 + A)$, $C_{out} = C(1 + 1/A)$. CE: $C_M = C_\mu(1 + g_m R_L')$.

f_H $\approx 1/2\pi R_{sig}'(C_\pi + C_M)$.

f_β $= 1/2\pi r_\pi(C_\pi + C_\mu)$; **$f_T$** $= \beta f_\beta = g_m/2\pi(C_\pi + C_\mu)$.

MOSFET: $g_m = 2K V_{ov} = 2I_D/V_{ov} = 2\sqrt{(KI_D)}$; $r_o = 1/\lambda I_D$.

CS loads: $-g_m(R_D \| r_o)$; $-g_m(r_{o1} \| r_{o2})$; $-g_{m1}/g_{m2} = -\sqrt{}$(ratio of W/L).

Cascade: gains multiply (dB add); $f_H(n) = f_H\sqrt{(2^{1/n} - 1)}$; $f_L(n) = f_L/\sqrt{(2^{1/n} - 1)}$.

Cascode: CE + CB; Miller ×2 only; $R_{out} \approx \beta r_o$ or $g_m r_o^2$; gain up to $(g_m r_o)^2$.

Practice — short questions (3 marks)

1. Draw the hybrid-π model of a BJT and define g_m, r_π, r_o.
2. What is the role of the bypass capacitor in a CE amplifier? What happens to gain if it is removed?
3. State Miller's theorem. An amplifier with gain −80 has C_{bc} = 3 pF; find C_{in}. [243 pF]
4. Define f_β and f_T and show $f_T = \beta f_\beta$.
5. Why does the overall bandwidth of a multistage amplifier decrease? Give the expression for $f_H(n)$.
6. Derive the small-signal resistance of a diode-connected MOSFET.
7. List three advantages of a cascode amplifier.

Practice — long questions (9/14 marks)

1. Draw an RC-coupled CE amplifier and its mid-band equivalent. Derive A_v, R_{in}, R_{out}. Evaluate for I_C = 2 mA, β = 150, R_C = 3.3 kΩ, R_L = 4.7 kΩ, R_B = 10 kΩ (ignore r_o). [g_m = 80 mS, r_π = 1.875 kΩ, $A_v \approx -155$, $R_{in} \approx 1.58$ kΩ]
2. Explain the low-frequency response of a CE amplifier and derive expressions for the poles due to C_{C1}, C_{C2} and C_E.
3. Derive the expression for the upper cut-off frequency of a CE amplifier using the Miller approximation.
4. Derive the short-circuit current gain of a BJT using the hybrid-π model. A BJT has f_T = 400 MHz at I_C = 2 mA, C_μ = 2 pF; find C_π. [≈ 29.8 pF]
5. Compare CS amplifiers with resistive, current-source and diode-connected loads with gain expressions.
6. Derive the output resistance of a MOS cascode. Why is a cascode preferred in wide-band amplifiers?

03

Feedback Amplifiers & Oscillators

> *Negative feedback is how every precise analog circuit is built; positive feedback is how every clock is born. Expect "four topologies" and "derive the frequency of oscillation" in every paper.*

SYLLABUS CHECKLIST

- ☐ Feedback concept; effect of negative feedback on gain, bandwidth, distortion, noise, input and output impedance
- ☐ Four topologies — voltage-series, voltage-shunt, current-series, current-shunt
- ☐ Analysis of voltage-series and voltage-shunt feedback amplifiers
- ☐ Oscillators — Barkhausen criterion
- ☐ RC phase-shift and Wien-bridge oscillators (with analysis)
- ☐ LC oscillators — Hartley and Colpitts
- ☐ Crystal oscillator — equivalent circuit, series and parallel resonance

Module 3 · CO3 · Analyse feedback amplifiers and design sinusoidal oscillators

3.1 The feedback idea

Feedback means taking a sample of the output and sending it back to the input. If the returned signal *opposes* the input it is **negative feedback**; if it *aids* the input it is **positive feedback**.

THINK OF IT LIKE

Driving a car. You look at where the car is going (sample the output) and correct the steering against the error (negative feedback). Without looking you would drift; with feedback, small bumps (component variations) hardly matter.

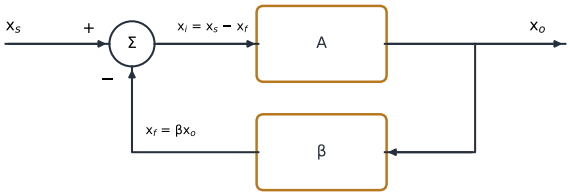


Fig 3.1 Basic negative-feedback loop. x may be a voltage or a current.

From Fig 3.1: $x_o = A\,x_i$, $x_f = \beta x_o$, $x_i = x_s - x_f$. Eliminate x_i: $x_o = A(x_s - \beta x_o)$, so

CLOSED-LOOP GAIN

$$A_f = \frac{x_o}{x_s} = \frac{A}{1 + A\beta} \qquad \text{loop gain} = A\beta, \quad \text{desensitivity factor } D = 1 + A\beta$$

If $A\beta \gg 1$: $A_f \approx \dfrac{1}{\beta}$ (set only by the feedback network — usually precise resistors)

KEY IDEA

Negative feedback **trades gain for quality**. You lose a factor $(1 + A\beta)$ in gain and get that same factor of improvement in almost everything else.

3.2 Effects of negative feedback

1. Gain stability (desensitivity)

Differentiate $A_f = A/(1 + A\beta)$ with respect to A: $dA_f/dA = 1/(1 + A\beta)^2$. Dividing by A_f:

$$\frac{dA_f}{A_f} = \frac{1}{1 + A\beta} \cdot \frac{dA}{A}$$

A 10 % change in A becomes a $10/(1 + A\beta)$ % change in A_f.

2. Bandwidth increases

Let $A(f) = A_m/(1 + jf/f_H)$. Then $A_f = A/(1 + A\beta) = [A_m/(1 + A_m\beta)] \,/\, [1 + jf/(f_H(1 + A_m\beta))]$. So

BANDWIDTH WITH FEEDBACK

$$f_{Hf} = f_H(1 + A_m\beta) \qquad f_{Lf} = \frac{f_L}{1 + A_m\beta} \qquad A_{mf} \times f_{Hf} = A_m \times f_H \text{ (GBW constant)}$$

3. Distortion and noise

Distortion produced *inside* the amplifier (e.g. in the output stage) is reduced to D/(1 + Aβ), because the feedback "sees" the distortion at the output and subtracts it at the input. The same holds for noise or hum generated *inside* the loop (after the input). Noise that enters together with the input signal is amplified just like the signal — feedback cannot remove it.

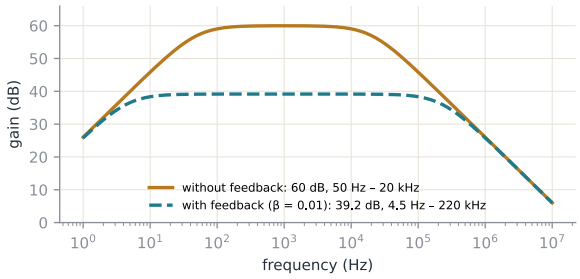


Fig 3.2 Feedback flattens and widens the response (A = 1000, β = 0.01, so 1 + Aβ = 11).

4. Input and output impedance

It depends on how the feedback is mixed at the input and sampled at the output — next section. Rule: **series** connection multiplies the impedance by (1 + Aβ); **shunt** connection divides it by (1 + Aβ).

SOLVED EXAMPLE 3.1 · DESENSITIVITY

An amplifier with A = 1000 has β = 0.01. Find A_f. If A drops by 20 %, by what percentage does A_f change?

- 1 + Aβ = 11 → A_f = 1000/11 = 90.9.
- New A = 800: A_f = 800/(1 + 8) = 88.9. Change = (90.9 − 88.9)/90.9 = 2.2 %.
- (Small-change formula: 20 %/11 = 1.8 % — close; the exact value is larger because 20 % is not "small".)

A_f = 90.9; it changes by only 2.2 %

SOLVED EXAMPLE 3.2 · BANDWIDTH AND DISTORTION

The same amplifier (A = 1000) has f_L = 50 Hz, f_H = 20 kHz and 10 % second-harmonic distortion. With β = 0.01, find the new cut-offs and distortion. What β gives A_f = 50 for an amplifier with A = 10^4?

- f_{Hf} = 20 k × 11 = 220 kHz; f_{Lf} = 50/11 = 4.55 Hz; distortion = 10/11 = 0.91 %.
- $A_f = A/(1 + A\beta) \rightarrow \beta = (A/A_f - 1)/A = (200 - 1)/10^4 = 0.0199$.

4.55 Hz – 220 kHz, 0.91 % distortion; β = 0.0199

3.3 The four feedback topologies

Two questions name every topology:

1. **What is sampled at the output?** Voltage (β network in *parallel* with the load, "shunt sampling") or current (β network in *series* with the load, "series sampling").
2. **How is it mixed at the input?** In series with the source (voltages add in a loop) or in shunt (currents add at a node).

The name is "*sampled quantity – mixing*": voltage-series, voltage-shunt, current-series, current-shunt (Fig 3.3).

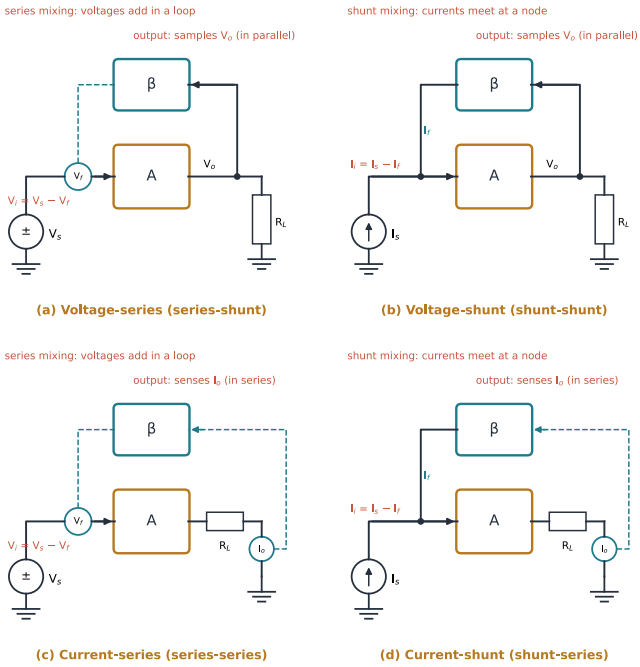


Fig 3.3 The four topologies (single-ended view, all blocks share a common ground).

Topology	A means	β means	Stabilises	R_{if}	R_{of}	Example
Voltage-series	$A_v = V_o/V_i$	V_f/V_o	voltage gain	R_iD ↑	R_o/D ↓	emitter follower, non-inverting op-amp
Voltage-shunt	$R_m = V_o/I_i$	I_f/V_o	transresistance	R_i/D ↓	R_o/D ↓	collector-feedback CE, inverting op-amp
Current-series	$G_m = I_o/V_i$	V_f/I_o	transconductance	R_iD ↑	R_oD ↑	CE with unbypassed R_E
Current-shunt	$A_i = I_o/I_i$	I_f/I_o	current gain	R_i/D ↓	R_oD ↑	two-stage CE pair with feedback from 2nd emitter

EXAM TIP

Identify a topology with the "short and open" test. **Output side:** short the load ($V_o = 0$). If feedback disappears → voltage sampling; if not → current sampling. **Input side:** if the feedback signal arrives at the *same node* as the input → shunt mixing; at a *different terminal* (e.g. emitter while input is at base) → series mixing.

3.4 Voltage-series feedback amplifier

The feedback network measures V_o and puts $V_f = \beta V_o$ in series with the input. The classic examples are the op-amp non-inverting amplifier ($\beta = R_1/(R_1 + R_f)$) and the emitter follower ($\beta = 1$, the whole output is fed back).

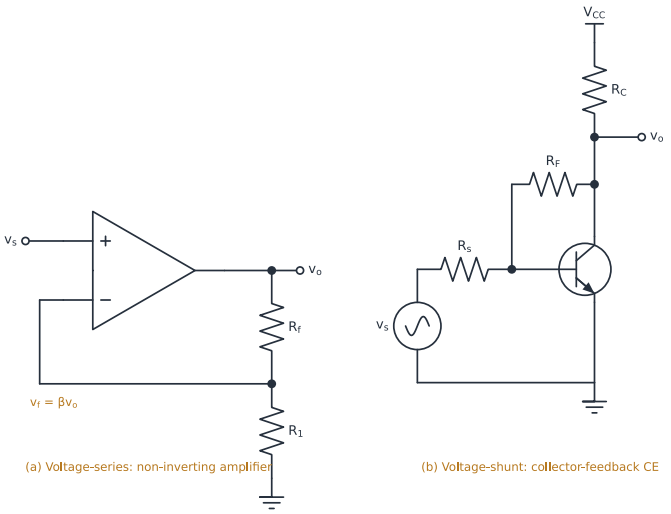


Fig 3.4 (a) Voltage-series and (b) voltage-shunt feedback amplifiers.

Model the basic amplifier as R_i at the input and a source AV_i behind R_o at the output (A = open-circuit voltage gain).

- **Gain:** $V_s = V_i + \beta V_o$ and $V_o = AV_i \rightarrow A_{vf} = A/(1 + A\beta)$.
- **Input resistance:** $V_s = V_i + \beta AV_i = V_i(1 + A\beta)$ and $I_i = V_i/R_i$, so $R_{if} = V_s/I_i = R_i(1 + A\beta)$. The feedback voltage opposes the source, so less current flows — the input looks bigger.
- **Output resistance:** set $V_s = 0$, drive the output with V. Then $V_i = -\beta V$ and $I = (V - AV_i)/R_o = V(1 + A\beta)/R_o$, so $R_{of} = R_o/(1 + A\beta)$. The amplifier fights any change in V_o — it looks like a stiffer voltage source.

VOLTAGE-SERIES

$$A_{vf} = \frac{A_v}{1 + A_v\beta} \qquad R_{if} = R_i(1 + A_v\beta) \qquad R_{of} = \frac{R_o}{1 + A_v\beta}$$

SOLVED EXAMPLE 3.3 · VOLTAGE-SERIES

An amplifier has $A_v = 500$, $R_i = 2\ k\Omega$, $R_o = 10\ k\Omega$. Voltage-series feedback with $\beta = 0.05$ is applied (e.g. $R_1 = 1\ k\Omega$, $R_f = 19\ k\Omega$). Find A_{vf}, R_{if}, R_{of}.

- $1 + A_v\beta = 1 + 25 = 26$.
- $A_{vf} = 500/26 = 19.2$ (ideal $1/\beta = 20$).
- $R_{if} = 2\ k \times 26 = 52\ k\Omega$; $R_{of} = 10\ k/26 = 385\ \Omega$.

$A_{vf} = 19.2$, $R_{if} = 52\ k\Omega$, $R_{of} = 385\ \Omega$

3.5 Voltage-shunt feedback amplifier

In Fig 3.4b the resistor R_F from collector to base samples the output voltage and feeds a current I_f back to the *base node* — the same node where the signal current enters. That is voltage sampling with shunt mixing. The natural gain here is the **transresistance** $R_m = V_o/I_i$, and $\beta = I_f/V_o = -1/R_F$ (sign: when V_o swings negative, current is pulled out of the base).

Convert the source to a Norton form ($I_s = V_s/R_s$ in parallel with R_s). Then $I_s = I_i + I_f$, $I_f = \beta V_o$, $V_o = R_m I_i$:

VOLTAGE-SHUNT

$$R_{mf} = \frac{R_m}{1 + \beta R_m} \qquad R_{if} = \frac{R_i}{1 + \beta R_m} \qquad R_{of} = \frac{R_o}{1 + \beta R_m} \qquad A_{vf} = \frac{V_o}{V_s} = \frac{R_{mf}}{R_s}$$

For large loop gain: $R_{mf} \to -R_F$ and $A_{vf} \to -R_F/R_s$ (the inverting-amplifier result!)

To include loading, the "basic amplifier without feedback" keeps R_F at both ends: at the input R_F appears in parallel with R_s and r_π (output shorted), at the output R_F appears in parallel with R_C (input shorted).

SOLVED EXAMPLE 3.4 · VOLTAGE-SHUNT (COLLECTOR FEEDBACK)

In Fig 3.4b, $R_s = 10\ k\Omega$, $R_F = 100\ k\Omega$, $R_C = 4.7\ k\Omega$, $g_m = 40\ mS$, $r_\pi = 2.5\ k\Omega$ (ignore r_o). Find R_{mf}, A_{vf}, R_{if} and R_{of}.

- Basic amplifier: $R_i = R_s \parallel R_F \parallel r_\pi = 10\ k \parallel 100\ k \parallel 2.5\ k = 1.96\ k\Omega$; $R_o = R_C \parallel R_F = 4.49\ k\Omega$.
- $R_m = V_o/I_s = -R_i\, g_m\, R_o = -1960.8 \times 0.04 \times 4489 = -352\ k\Omega$.
- $\beta = -1/R_F = -10^{-5}\ S \to 1 + \beta R_m = 1 + 3.52 = 4.52$.
- $R_{mf} = -352/4.52 = -77.9\ k\Omega \to A_{vf} = R_{mf}/R_s = -7.79$ (ideal $-R_F/R_s = -10$; the loop gain is only 3.5).
- $R_{if} = 1.96\ k/4.52 = 434\ \Omega$; $R_{of} = 4.49\ k/4.52 = 993\ \Omega$.

$R_{mf} = -77.9\ k\Omega$, $A_{vf} = -7.79$, $R_{if} = 434\ \Omega$, $R_{of} = 993\ \Omega$

INTERVIEW CORNER

Q. Which topology would you use for an ideal voltage amplifier, and why?

A. Voltage-series: it raises R_{in} (does not load the source) and lowers R_{out} (drives any load) — exactly what an ideal voltage amplifier needs.

Q. Can negative feedback make an amplifier unstable?

A. Yes. At high frequency extra phase shift can turn negative feedback into positive. If $|A\beta| \geq 1$ where the total phase reaches 180°, it oscillates — hence phase margin and compensation.

3.6 Oscillators and the Barkhausen criterion

An **oscillator** is an amplifier that feeds its own input — with **positive feedback** — so it produces a steady sine wave with no input at all. With positive feedback the loop equation becomes $A_f = A/(1 - A\beta)$. If $A\beta = 1$, $A_f \to \infty$: an output exists with zero input.

BARKHAUSEN CRITERION

$$|A\beta| = 1 \quad \text{and} \quad \angle A\beta = 0° \text{ (or } 360°) \quad \text{— at exactly one frequency } f_0$$

How does it start? There is always electrical noise (all frequencies). The frequency-selective β network picks f_0. If we design $|A\beta|$ slightly **greater than 1** at start-up, that component grows each trip round the loop. As the amplitude rises the amplifier begins to saturate, its effective gain drops, and the amplitude settles where $|A\beta| = 1$ exactly (Fig 3.5).

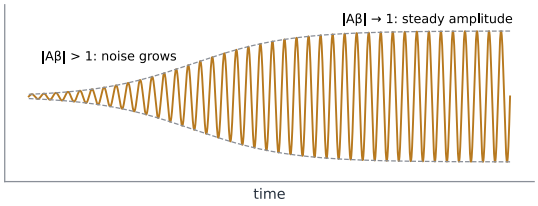


Fig 3.5 Oscillation building up from noise and settling.

3.7 RC phase-shift oscillator

An inverting amplifier gives 180°. We need another 180° from the feedback network. One RC section can give at most 90° (and only at zero gain), so we use **three RC sections**, each giving about 60° at f_0 (Fig 3.6).

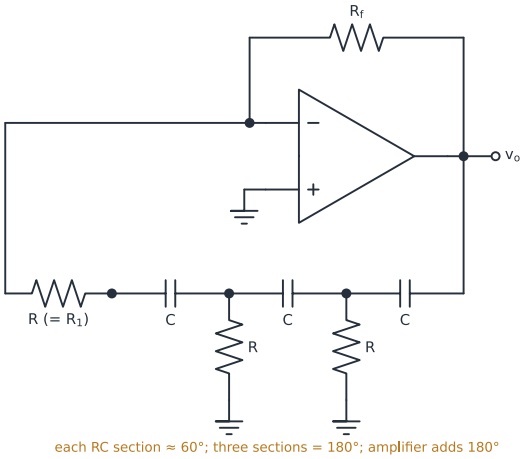


Fig 3.6 RC phase-shift oscillator with an op-amp inverting amplifier (gain $-R_f/R$). The last R is the amplifier's input resistor, returned to virtual ground.

Analysis

Write mesh equations for the three-section ladder (C in series, R to ground) driven by V_o, with currents I_1, I_2, I_3 and $X = 1/(\omega C)$. Solving for the voltage across the last R ($V_f = I_3 R$) gives, with $\alpha = 1/(\omega RC)$:

$$\beta = \frac{V_f}{V_o} = \frac{1}{(1 - 5\alpha^2) - j(6\alpha - \alpha^3)}$$

For β to be real (total phase exactly 180°) the imaginary part must vanish: $6\alpha - \alpha^3 = 0 \rightarrow \alpha^2 = 6 \rightarrow \omega = 1/(\sqrt{6}\,RC)$. At that frequency $\beta = 1/(1 - 30) = -1/29$. Barkhausen $|A\beta| = 1$ then needs $|A| = 29$.

RC PHASE-SHIFT OSCILLATOR

$$f_0 = \frac{1}{2\pi RC\sqrt{6}} \qquad \beta = -\frac{1}{29} \qquad |A| \geq 29 \quad \text{(op-amp: } R_f \geq 29R\text{)}$$

Transistor version (collector resistor R_C): $f_0 = \frac{1}{2\pi RC\sqrt{(6 + 4R_C/R)}}$, $h_{fe} \geq 23 + 29\frac{R}{R_C} + 4\frac{R_C}{R}$

SOLVED EXAMPLE 3.5 · RC PHASE-SHIFT

Design an op-amp RC phase-shift oscillator with R = 10 kΩ and C = 10 nF. Find f_0 and the minimum R_f.

- $f_0 = 1/(2\pi \times 10^4 \times 10^{-8} \times 2.449) = 650$ Hz.
- $|A| = R_f/R \geq 29 \rightarrow R_f \geq 290$ kΩ (use a little more, e.g. a 330 kΩ or a trimmer, to ensure start-up).

$f_0 \approx 650$ Hz, $R_f \geq 290$ kΩ

3.8 Wien-bridge oscillator

Here the amplifier is **non-inverting** (0°), so the feedback network must give 0° at f_0. A **series RC** followed by a **parallel RC** (a "lead-lag" network) does exactly that: at low frequency the series C blocks, at high frequency the parallel C shorts, and in between there is one frequency where the phase is zero.

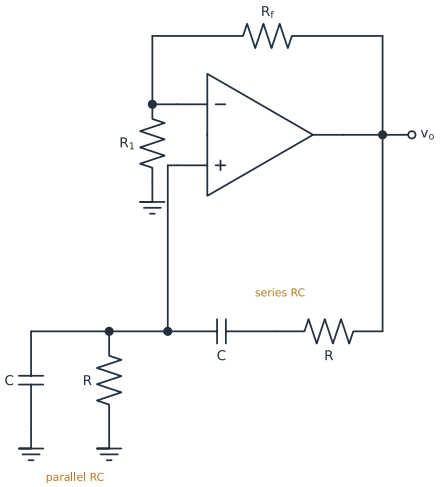


Fig 3.7 Wien-bridge oscillator: positive feedback through the RC lead-lag arms, negative feedback through R_f–R_1.

Analysis

$Z_s = R + 1/(j\omega C)$, $Z_p = R/(1 + j\omega RC)$. Then

$$\beta = \frac{Z_p}{Z_s + Z_p} = \frac{1}{3 + j(\omega RC - 1/\omega RC)}$$

β is real when $\omega RC = 1/(\omega RC)$, i.e. $\omega_0 = 1/RC$; there $\beta = 1/3$. The non-inverting gain $A = 1 + R_f/R_1$ must be at least 3.

WIEN BRIDGE

$$f_0 = \frac{1}{2\pi RC} \qquad \beta = 1/3 \qquad A = 1 + R_f/R_1 \geq 3 \;\Rightarrow\; R_f \geq 2R_1$$

Amplitude control: a lamp or thermistor in the R_1/R_f arm, or back-to-back diodes across part of R_f, lowers the gain automatically as amplitude rises — giving a very clean sine. Wien bridges are the standard audio-frequency (10 Hz–1 MHz) oscillator.

SOLVED EXAMPLE 3.6 · WIEN BRIDGE

$R = 10\ k\Omega$, $C = 0.01\ \mu F$, $R_1 = 10\ k\Omega$. Find f_0 and the required R_f. Verify the phase is zero at f_0.

- $f_0 = 1/(2\pi \times 10^4 \times 10^{-8}) = 1.59$ kHz.
- At ω_0: $\omega RC = 1$, so the j-term vanishes → $\beta = 1/3$, phase 0° √.
- $R_f \geq 2 \times 10\ k = 20\ k\Omega$ (slightly more for reliable start-up).

$f_0 = 1.59$ kHz, $R_f \geq 20\ k\Omega$

3.9 LC oscillators — Hartley and Colpitts

Above about 1 MHz, RC values become awkwardly small, so we use an **LC tank**. Energy sloshes between the inductor's magnetic field and the capacitor's electric field — like a pendulum — at $f_0 = 1/(2\pi\sqrt{LC})$. The transistor just tops up the energy lost in each cycle.

Both circuits use the same skeleton (Fig 3.8): reactance X_1 across the output (collector–emitter), X_2 across the input (base–emitter), and X_3 bridging them. Oscillation needs $X_1 + X_2 + X_3 = 0$, with X_1, X_2 of the same type and X_3 of the opposite type. The feedback fraction is $\beta = -X_2/X_1$, which together with the inverting CE stage gives the required 360°.

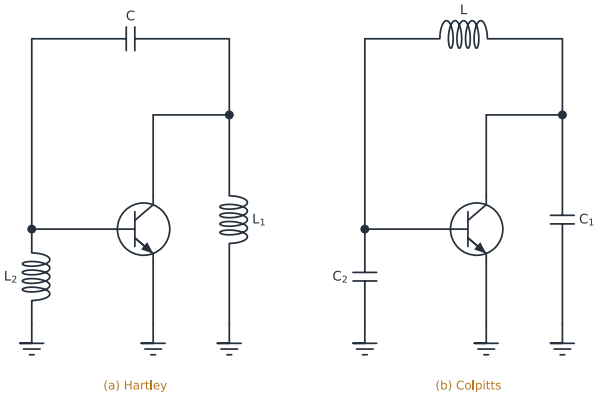


Fig 3.8 Hartley (tapped inductor) and Colpitts (tapped capacitor) oscillators — AC skeleton; biasing, RFC and coupling capacitors omitted.

	Hartley	Colpitts
Tank	L_1, L_2 (tapped coil) + C	C_1, C_2 (tapped capacitor) + L
Frequency	$f_0 = \frac{1}{2\pi\sqrt{((L_1 + L_2 + 2M)C)}}$	$f_0 = \frac{1}{2\pi\sqrt{(L\ C_{eq})}}$, $C_{eq} = \frac{C_1C_2}{C_1 + C_2}$
Feedback fraction $\lvert\beta\rvert$	$(L_2 + M)/(L_1 + M)$	C_1/C_2
Gain condition	h_{fe} (or $\lvert A\rvert$) $\geq L_1/L_2$ ($M = 0$)	h_{fe} (or $\lvert A\rvert$) $\geq C_2/C_1$
Notes	Easy to tune with one variable C	Better purity at high f; capacitors cleaner than tapped coils

SOLVED EXAMPLE 3.7 · HARTLEY AND COLPITTS

(a) Hartley: L_1 = 1 mH, L_2 = 0.1 mH (M = 0), C = 200 pF. (b) Colpitts: C_1 = 1 nF, C_2 = 10 nF, L = 100 µH. Find f_0 and the minimum gain for each.

- (a) L_{eq} = 1.1 mH → $f_0 = 1/(2\pi\sqrt{(1.1 \times 10^{-3} \times 200 \times 10^{-12})})$ = 339 kHz; $|A| \geq L_1/L_2$ = 10.
- (b) C_{eq} = (1 × 10)/(11) nF = 0.909 nF → $f_0 = 1/(2\pi\sqrt{(100\ \mu \times 0.909\ n)})$ = 528 kHz; $|A| \geq C_2/C_1$ = 10.

(a) 339 kHz, A ≥ 10 (b) 528 kHz, A ≥ 10

3.10 Crystal oscillator

LC oscillators drift: L and C change with temperature and ageing. A **quartz crystal** is a tiny slab that vibrates mechanically at an extremely precise frequency. Because quartz is **piezoelectric** (squeeze it → voltage; apply voltage → it bends), that mechanical resonance shows up electrically as a resonant circuit with an enormous Q (10^4–10^6).

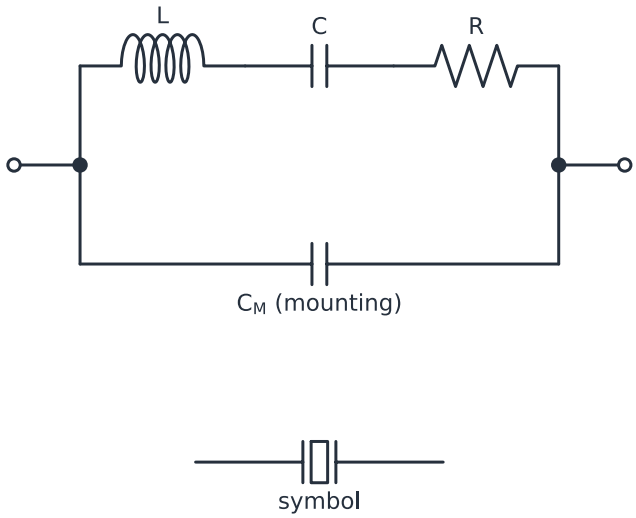


Fig 3.9 Crystal equivalent circuit: L (mass), C (compliance), R (friction) and the electrode capacitance C_M.

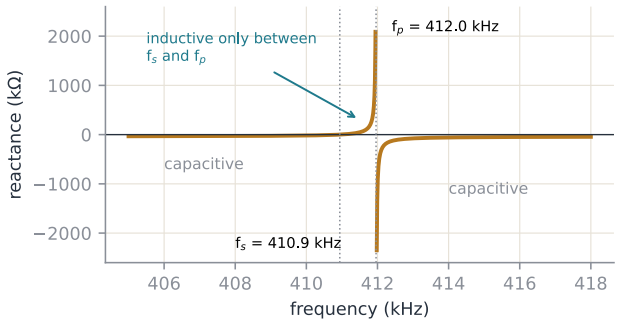


Fig 3.10 Reactance vs frequency (L = 3 H, C = 0.05 pF, C_M = 10 pF).

SERIES AND PARALLEL RESONANCE

Series (L–C branch alone): $f_s = \dfrac{1}{2\pi\sqrt{(LC)}}$ — impedance minimum (≈ R), purely resistive

Parallel (branch with C_M): $f_p = \dfrac{1}{2\pi\sqrt{(L\,C_{eq})}}$, $C_{eq} = \dfrac{C\,C_M}{C + C_M}$ — impedance maximum; $f_p = f_s\sqrt{(1 + C/C_M)} > f_s$

Because $C \ll C_M$, f_p is only slightly above f_s. Between them the crystal is **inductive**; everywhere else it is capacitive. Two ways to use it:

- **Series mode:** crystal in the feedback path; it passes the signal only near f_s (low impedance).
- **Parallel mode:** crystal replaces the inductor of a Colpitts (the **Pierce oscillator**, used in every microcontroller clock). It operates between f_s and f_p, where it looks like a very high-Q inductor.

SOLVED EXAMPLE 3.8 · CRYSTAL

A crystal has L = 3 H, C = 0.05 pF, R = 2 kΩ, C_M = 10 pF. Find f_s, f_p, their separation and the Q.

- $f_s = 1/(2\pi\sqrt{(3 \times 0.05 \times 10^{-12})}) = 410.94$ kHz.
- $C_{eq} = 0.05 \times 10/10.05 = 0.04975$ pF → $f_p = 411.96$ kHz ($= f_s\sqrt{1.005}$).
- Separation ≈ 1.03 kHz (0.25 %). $Q = \omega_s L/R = 2\pi \times 410.94\text{ k} \times 3/2000 \approx 3870$.

f_s = 410.9 kHz, f_p = 412.0 kHz, $\Delta f \approx 1.03$ kHz, Q ≈ 3870

✓ Module 3 — Quick revision

Feedback: $A_f = A/(1 + A\beta)$; $A\beta \gg 1 \rightarrow A_f \approx 1/\beta$.

Sensitivity: $dA_f/A_f = (dA/A)/(1 + A\beta)$.

Bandwidth: $f_{Hf} = f_H(1 + A\beta)$, $f_{Lf} = f_L/(1 + A\beta)$; GBW constant.

Distortion & internal noise: ÷ $(1 + A\beta)$.

Series mixing: R_{in} × D. **Shunt mixing:** R_{in} ÷ D.

Voltage sampling: R_{out} ÷ D. **Current sampling:** R_{out} × D.

Voltage-series: stabilises A_v (non-inverting op-amp, emitter follower).

Voltage-shunt: $\beta = -1/R_F$; $A_{vf} \rightarrow -R_F/R_s$.

Barkhausen: $|A\beta| = 1$, $\angle A\beta = 0°/360°$. Start with $|A\beta|$ slightly > 1.

RC phase-shift: $f = 1/(2\pi RC\sqrt{6})$, $\beta = -1/29$, $|A| \geq 29$.

Wien: $f = 1/2\pi RC$, $\beta = 1/3$, $A \geq 3$ ($R_f \geq 2R_1$).

Hartley: $f = 1/2\pi\sqrt{((L_1+L_2+2M)C)}$, $A \geq L_1/L_2$.

Colpitts: $f = 1/2\pi\sqrt{(LC_1C_2/(C_1+C_2))}$, $A \geq C_2/C_1$.

Crystal: $f_s = 1/2\pi\sqrt{(LC)}$; $f_p = f_s\sqrt{(1 + C/C_M)}$; inductive between them.

Practice — short questions (3 marks)

1. Show that negative feedback increases bandwidth by $(1 + A\beta)$.
2. An amplifier has A = 2000 and $\beta = 0.05$. Find A_f and the % change in A_f for a 10 % change in A. [19.8; ≈ 0.1 %]
3. Identify the topology of an emitter follower and justify.
4. State the Barkhausen criterion. Why must $|A\beta|$ exceed 1 at start-up?
5. Why is an LC oscillator preferred over an RC oscillator at radio frequencies?
6. Draw the equivalent circuit of a crystal and define series and parallel resonance.
7. Find f_0 of a Colpitts oscillator with L = 50 µH, $C_1 = C_2 = 2$ nF. [≈ 712 kHz]

Practice — long questions (9/14 marks)

1. Explain the four feedback topologies with block diagrams. Tabulate their effect on R_{in} and R_{out}.
2. Derive R_{if} and R_{of} of a voltage-series feedback amplifier.
3. Analyse a voltage-shunt (collector-feedback) amplifier and obtain R_{mf}, R_{if}, R_{of}.
4. Derive the frequency of oscillation and the gain condition of an RC phase-shift oscillator. Design one for 1 kHz with C = 10 nF. [R ≈ 6.5 kΩ, $R_f \geq$ 189 kΩ]
5. Derive f_0 and the gain requirement for a Wien-bridge oscillator. How is amplitude stabilised?
6. Explain the working of Hartley and Colpitts oscillators with circuit diagrams and derive their frequency of oscillation.

04

Power Amplifiers & Voltage Regulators

> *Every speaker driver and every power rail on a PCB uses these ideas. Interviewers love "why is class B only 78.5 % efficient?" and "how does fold-back protect a regulator?"*

SYLLABUS CHECKLIST

- ☐ Power amplifiers — classification (A, B, AB, C)
- ☐ Class A transformer-coupled amplifier — efficiency derivation
- ☐ Class B push-pull and complementary-symmetry — efficiency, crossover distortion
- ☐ Class AB amplifier
- ☐ Linear voltage regulators — Zener shunt regulator, transistor series regulator, shunt regulator
- ☐ Short-circuit and fold-back protection
- ☐ Current boosting with an external pass transistor

Module 4 · CO4 · Analyse power amplifiers for efficiency and design linear regulated supplies

4.1 Classification of power amplifiers

The amplifiers of Module 2 were *small-signal*: tiny swings, and we cared about gain. A **power amplifier** is the last stage that drives a real load (a speaker, motor, antenna). Swings are large, so the questions change: **how much power reaches the load, how much is wasted as heat, and how distorted is the output?**

CONVERSION EFFICIENCY

$$\eta = \frac{P_{ac}\ (\text{signal power delivered to load})}{P_{dc}\ (\text{power drawn from supply})} \times 100\ \% \qquad P_D\ (\text{transistor heat}) = P_{dc} - P_{ac}$$

Power amplifiers are classified by the **conduction angle** — for how much of each input cycle the transistor carries current (Fig 4.1).

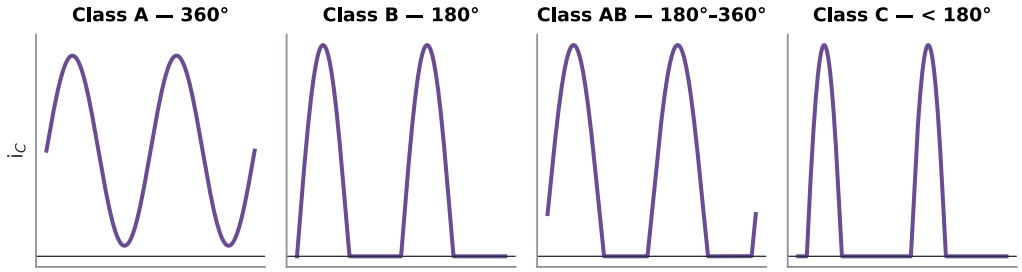


Fig 4.1 Collector current for each class with a sine input.

Class	Conduction	Q-point	Max η	Distortion	Use
A	360°	middle of load line	25 % (resistive load), 50 % (transformer)	lowest	small power, drivers
B	180°	at cut-off	78.5 %	crossover	push-pull audio
AB	180°–360°	just above cut-off	between A and B	low	almost all audio amplifiers
C	< 180°	below cut-off	up to ~90 %+	very high (needs tuned load)	RF transmitters

4.2 Class A transformer-coupled amplifier

In a plain (series-fed) class A stage the quiescent current flows through the load resistor itself, wasting DC power, and the maximum efficiency is only 25 %. Putting the load on the secondary of a transformer (Fig 4.2) fixes two problems at once:

- The primary winding has almost zero DC resistance, so no DC power is wasted in the load path.
- The transformer **matches impedance**: a small load (say an 8 Ω speaker) looks like a larger resistance on the collector side:

$$R_L' = n^2 R_L, \quad n = N_1/N_2$$

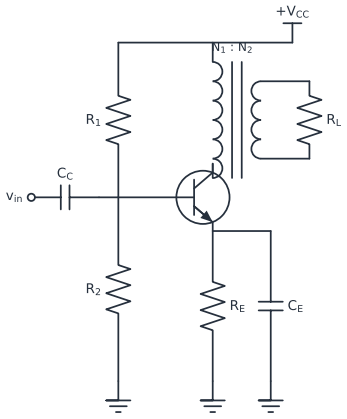


Fig 4.2 Transformer-coupled class A power amplifier.

Load lines

Because the primary's DC resistance ≈ 0, the **DC load line is vertical** at $V_{CE} = V_{CC}$: $V_{CEQ} \approx V_{CC}$. For signals the collector sees R_L', so the **AC load line** passes through Q with slope $-1/R_L'$ (Fig 4.3). The transformer's stored energy lets the collector swing above V_{CC} — up to $2V_{CC}$.

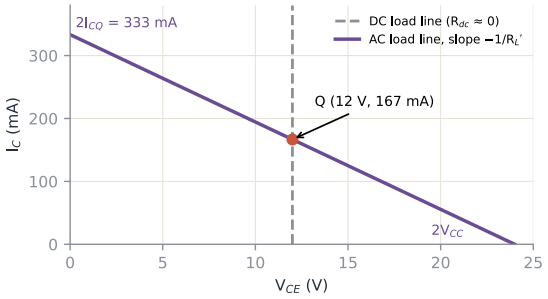


Fig 4.3 DC and AC load lines for Example 4.1 (V_{CC} = 12 V, R_L' = 72 Ω). Q is placed at the centre of the AC load line.

Efficiency derivation

For maximum symmetric swing place Q at the centre of the AC load line: $V_{CEQ} = V_{CC}$ and $I_{CQ} = V_{CC}/R_L'$. The collector voltage can then swing from 0 to $2V_{CC}$ (peak $V_p = V_{CC}$) and the current from 0 to $2I_{CQ}$ (peak $I_p = I_{CQ}$).

CLASS A (TRANSFORMER-COUPLED)

$$P_{dc} = V_{CC}\, I_{CQ} \quad \text{(constant, signal or no signal)}$$

$$P_{ac} = \frac{V_p I_p}{2} = \frac{V_p^{\,2}}{2R_L'} \quad \rightarrow \quad P_{ac(max)} = \frac{V_{CC} I_{CQ}}{2}$$

$$\eta_{max} = \frac{V_{CC} I_{CQ}/2}{V_{CC} I_{CQ}} = 50\ \%$$

The transistor dissipates the most when there is **no signal** ($P_D = P_{dc}$), so the heat sink must handle the full P_{dc}. Real efficiency is lower (30–40 %) because the swing cannot go fully to 0 V.

SOLVED EXAMPLE 4.1 · CLASS A TRANSFORMER-COUPLED

An 8 Ω speaker is driven through a 3 : 1 transformer from V_{CC} = 12 V. Find R_L', the I_{CQ} for maximum swing, P_{dc}, maximum P_{ac} and η. What are P_{ac} and η if the collector swing is only 10 V peak?

- $R_L' = 3^2 \times 8 = 72\ \Omega$; $I_{CQ} = 12/72 = 166.7$ mA.
- $P_{dc} = 12 \times 0.1667 = 2.0$ W; $P_{ac(max)} = 2.0/2 = 1.0$ W; $\eta = 50$ %.
- $V_p = 10$ V: $P_{ac} = 10^2/(2 \times 72) = 0.694$ W; $\eta = 0.694/2 = 34.7$ %; $P_D = 1.31$ W.

$R_L' = 72\ \Omega$, $I_{CQ} = 167$ mA, $P_{dc} = 2$ W, $P_{ac(max)} = 1$ W (50 %); at 10 V peak: 0.69 W, 34.7 %

4.3 Class B push-pull and complementary symmetry

A class B transistor is biased exactly at cut-off, so it conducts for only half the cycle and draws **no current with no signal**. To rebuild the full sine we use **two** transistors in **push-pull**: one handles the positive half, the other the negative half.

Transformer-coupled push-pull

The centre-tapped input transformer gives Q_1 and Q_2 base signals 180° apart. On the positive half Q_1 conducts and pushes current through the top half of the output primary; on the negative half Q_2 pulls current through the bottom half. The output transformer adds the two halves into a full sine. A bonus: **even harmonics cancel** in push-pull, and the DC currents of the two halves cancel in the core (no saturation).

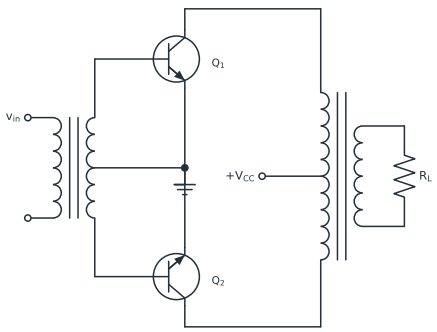


Fig 4.4 Class B push-pull amplifier with centre-tapped input and output transformers.

Complementary-symmetry (transformer-less) push-pull

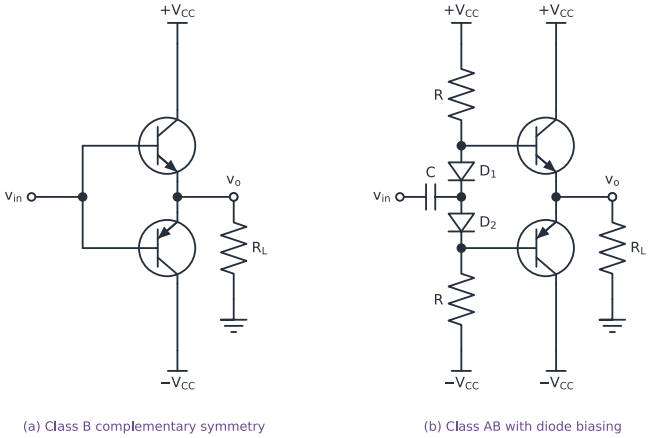


Fig 4.5 (a) Class B complementary-symmetry stage; (b) class AB version with diode biasing.

An NPN and a matched PNP are connected as emitter followers with $\pm V_{CC}$ supplies (Fig 4.5a). Positive input → NPN conducts (sourcing current into R_L); negative input → PNP conducts (sinking current). No bulky transformers, no phase splitter, low output resistance — this is the standard audio output stage. Its cost is the need for a matched complementary pair and two supplies.

Efficiency derivation

Let the output be $v_o = V_p \sin \omega t$ across R_L (R_L' for the transformer version), with $I_p = V_p/R_L$. Each supply delivers a half-wave rectified current, whose average is I_p/π. Two such halves flow from V_{CC} (or from $\pm V_{CC}$):

CLASS B

$$P_{dc} = V_{CC} \cdot \frac{2I_p}{\pi} \qquad P_{ac} = \frac{V_p I_p}{2} = \frac{V_p^2}{2R_L}$$

$$\eta = \frac{V_p I_p/2}{2V_{CC} I_p/\pi} = \frac{\pi}{4} \cdot \frac{V_p}{V_{CC}} \quad \rightarrow \quad \eta_{max} = \pi/4 = 78.5\ \% \quad (V_p = V_{CC})$$

$$P_D\ (\text{both}) = \frac{2V_{CC}V_p}{\pi R_L} - \frac{V_p^2}{2R_L}; \quad dP_D/dV_p = 0 \rightarrow V_p = \frac{2V_{CC}}{\pi}, \quad P_{D(max)} = \frac{2V_{CC}^2}{\pi^2 R_L} \approx 0.4\ P_{ac(max)}$$

SOLVED EXAMPLE 4.2 · CLASS B COMPLEMENTARY SYMMETRY

A class B complementary stage uses ±12 V and an 8 Ω load. For an output of 10 V peak find P_{ac}, P_{dc}, η and the dissipation per transistor. Also find the maximum possible P_{ac} and the worst-case dissipation per transistor.

- $I_p = 10/8 = 1.25$ A; $P_{ac} = 10 \times 1.25/2 = 6.25$ W.
- $P_{dc} = 12 \times 2 \times 1.25/\pi = 9.55$ W → $\eta = 6.25/9.55 = 65.4\ \%$ (check: $\pi/4 \times 10/12 = 65.4\ \%$ √).
- $P_D = 9.55 - 6.25 = 3.30$ W total → 1.65 W per transistor.
- $P_{ac(max)} = 12^2/(2 \times 8) = 9$ W. $P_{D(max)} = 2 \times 144/(\pi^2 \times 8) = 3.65$ W total (at $V_p = 7.64$ V) → 1.82 W per transistor.

6.25 W, 9.55 W, 65.4 %, 1.65 W each; $P_{ac(max)}$ = 9 W, worst case 1.82 W each

Crossover distortion

A silicon transistor does not conduct until $V_{BE} \approx 0.6$–0.7 V. With the bases tied together and no bias (pure class B), for $-0.7\text{ V} < v_{in} < +0.7\text{ V}$ **neither** transistor conducts, and the output sits at zero. This "dead zone" each time the signal crosses zero is **crossover distortion** (Fig 4.7). It is most noticeable at low volume, which is exactly when the ear is most sensitive to it.

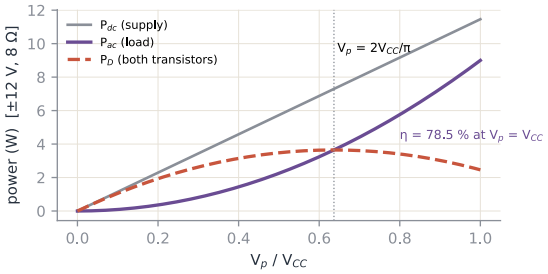


Fig 4.6 Class B powers vs output amplitude (±12 V, 8 Ω). Transistor heat peaks at $V_p \approx 0.64\ V_{CC}$, not at full output.

4.4 Class AB and class C

Class AB

Give each transistor a small forward bias, just enough to sit at the edge of conduction ($V_{BE} \approx 0.6$–0.7 V) with a small quiescent current. Then the moment the signal moves, one transistor is already conducting — the dead zone disappears. Each device now conducts for slightly more than 180°.

- **Diode biasing** (Fig 4.5b): two diodes D_1, D_2 between the bases drop $\approx 2V_{BE}$. If the diodes are thermally coupled to the transistors (same heat sink), their drop falls with temperature just as V_{BE} does, preventing thermal runaway. This is the **current-mirror** idea.
- **V_{BE} multiplier**: a transistor with a resistor divider across its base gives an adjustable bias voltage $V_{BE}(1 + R_a/R_b)$ — used in hi-fi amplifiers to trim quiescent current.

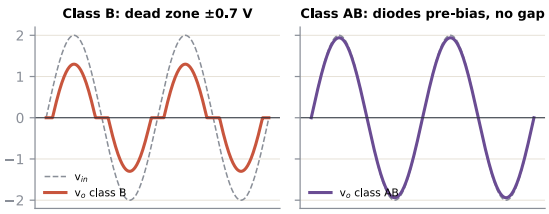


Fig 4.7 Crossover distortion in class B and its removal in class AB (2 V peak input).

Efficiency is a little below class B (the idle current wastes some power) but distortion is far lower — which is why nearly every audio power amplifier is class AB.

Class C (for completeness)

Biased below cut-off, so the transistor conducts in short pulses (< 180°). The output is heavily distorted, but a **tuned LC tank** at the collector "rings" at the fundamental and restores a clean sine. Efficiency can exceed 90 %. Used only for single-frequency RF power (transmitters, frequency multipliers) — never for audio.

INTERVIEW CORNER

Q. Why is class B efficiency limited to 78.5 %?

A. The supply delivers half-sine current pulses with average $2I_p/\pi$ at the full V_{CC}, while the load gets $V_pI_p/2$. Even when $V_p = V_{CC}$, the ratio is $\pi/4$: the transistor always drops part of V_{CC} during the cycle.

Q. When does a class B transistor run hottest?

A. At $V_p = 2V_{CC}/\pi$ (≈ 64 % of full swing), not at full power. Heat sinks are sized for $P_{D(max)} = 2V_{CC}^2/\pi^2R_L$ (both devices).

Q. Why are the bias diodes in class AB mounted on the heat sink?

A. So their forward voltage tracks the transistors' V_{BE} with temperature. Otherwise, as the transistors heat up, their V_{BE} falls, quiescent current rises and the stage can run away thermally.

4.5 Linear voltage regulators — the basics

A rectifier-filter gives a DC voltage that sags under load, follows mains fluctuations and carries ripple. A **voltage regulator** holds the output constant regardless. A *linear* regulator does this by placing a control element (Zener or transistor) that behaves like a self-adjusting resistor, burning off the excess voltage as heat.

Every linear regulator has the same blocks: a **reference** (usually a Zener), a **sampling/comparison** circuit, and a **control element** — either in **series** with the load (series regulator) or in **parallel** with it (shunt regulator).

PERFORMANCE MEASURES

$$\text{Line regulation} = \frac{\Delta V_o}{\Delta V_{in}} \text{ (load fixed)} \qquad \text{Load regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\ \% \qquad R_o = -\frac{\Delta V_o}{\Delta I_L}$$

Ideal regulator: line regulation = 0, load regulation = 0 %, $R_o = 0$. Ripple rejection (in dB) is a third common figure.

4.6 Zener shunt regulator

A Zener diode in reverse breakdown keeps an almost constant voltage V_Z across itself over a wide range of current. Put it across the load with a series resistor R_S (Fig 4.8). The current through R_S is roughly fixed; it **shares** between Zener and load. If the load takes more, the Zener takes less, and V_o stays at V_Z.

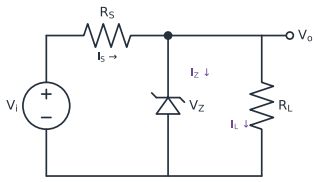


Fig 4.8 Zener shunt regulator.

ZENER REGULATOR

$$I_S = \frac{V_i - V_Z}{R_S}, \quad I_L = \frac{V_Z}{R_L}, \quad I_Z = I_S - I_L, \quad P_Z = V_Z I_Z$$

Zener is ON only if the open-circuit voltage $V_i R_L/(R_S + R_L) \geq V_Z \Rightarrow R_{L(min)} = \dfrac{R_S V_Z}{V_i - V_Z}$

Design: $R_{S(max)} = \dfrac{V_{i(min)} - V_Z}{I_{L(max)} + I_{Z(min)}}$; check $I_{Z(max)} = \dfrac{V_{i(max)} - V_Z}{R_S} - I_{L(min)}$ and $P_{Z(max)}$

SOLVED EXAMPLE 4.3 · IS THE ZENER ON?

V_i = 16 V, R_S = 1 kΩ, V_Z = 10 V. Find V_L, I_Z and P_Z for (a) R_L = 1.2 kΩ, (b) R_L = 3 kΩ. What is the smallest R_L that keeps the Zener on?

▸ (a) Remove the Zener: V = 16 × 1.2/2.2 = 8.73 V < 10 V → Zener OFF. V_L = 8.73 V, I_Z = 0. (Not regulating!)

▸ (b) V = 16 × 3/4 = 12 V > 10 V → Zener ON, V_L = 10 V. I_S = 6/1 k = 6 mA, I_L = 10/3 k = 3.33 mA, I_Z = 2.67 mA, P_Z = 26.7 mW.

▸ $R_{L(min)}$ = 1 k × 10/6 = 1.67 kΩ.

(a) 8.73 V, Zener off; (b) 10 V, I_Z = 2.67 mA, P_Z = 26.7 mW; $R_{L(min)}$ = 1.67 kΩ

SOLVED EXAMPLE 4.4 · ZENER REGULATOR DESIGN

Design a 12 V Zener regulator for an input of 18–24 V and load current 0–50 mA. Take $I_{Z(min)}$ = 5 mA. Find R_S and the Zener power rating.

▸ Worst case for keeping regulation: lowest V_i, highest I_L. $R_{S(max)}$ = (18 − 12)/(50 + 5) mA = 109 Ω → choose 100 Ω.

▸ Worst case for Zener heating: highest V_i, no load. $I_{Z(max)}$ = (24 − 12)/100 = 120 mA → P_Z = 12 × 0.12 = 1.44 W → use a 2 W Zener.

R_S = 100 Ω; Zener ≥ 1.44 W (use 2 W)

COMMON MISTAKE

Assuming the Zener is always conducting. Always check first by removing it and computing the open-circuit voltage across its terminals.

4.7 Transistor series regulator

A Zener alone can supply only small load currents (its own current must swing as much as the load's). Add a transistor as a **current amplifier**: the Zener holds the base steady, and the transistor — an emitter follower in series with the load — supplies the big current (Fig 4.9a).

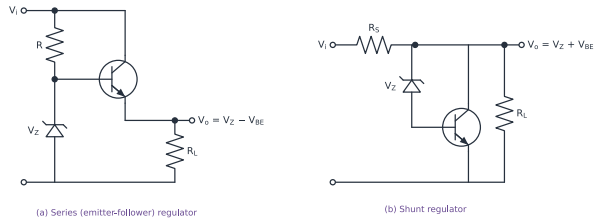


Fig 4.9 (a) Emitter-follower series regulator; (b) transistor shunt regulator.

How it regulates: $V_{BE} = V_Z - V_o$. If V_o tries to fall (more load), V_{BE} rises, the transistor conducts harder and restores V_o. If V_i rises, the extra voltage simply appears across the transistor's V_{CE}.

SERIES REGULATOR

$$V_o = V_Z - V_{BE} \quad I_B = \frac{I_L}{\beta + 1} \quad I_R = \frac{V_i - V_Z}{R} \quad I_Z = I_R - I_B \quad P_D = (V_i - V_o)I_C$$

With error amplifier (feedback series regulator): $V_o = \frac{R_1 + R_2}{R_2}(V_Z + V_{BE2})$ — output adjustable and much better regulated

In the improved version, a second transistor (or op-amp) compares a fraction of V_o (from divider R_1–R_2) with the Zener reference and drives the pass transistor's base — negative feedback (Module 3!) makes the output resistance tiny.

SOLVED EXAMPLE 4.5 · SERIES REGULATOR

In Fig 4.9a, V_i = 20 V, R = 1 kΩ, V_Z = 12 V, β = 50, R_L = 100 Ω. Find V_o, I_L, I_Z and the pass-transistor dissipation. For a feedback regulator with V_Z = 6.2 V and $R_1 = R_2$ = 10 kΩ, what is V_o?

- $V_o = 12 - 0.7 = 11.3$ V; $I_L = 11.3/100 = 113$ mA.
- $I_B = 113/51 = 2.22$ mA; $I_R = (20 - 12)/1\text{ k} = 8$ mA → $I_Z = 8 - 2.22 = 5.78$ mA.
- $I_C = 50 \times 2.22 = 110.8$ mA → $P_D = (20 - 11.3) \times 0.1108 = 0.96$ W.
- Feedback version: $V_o = (20\text{ k}/10\text{ k})(6.2 + 0.7) = 13.8$ V.

11.3 V, 113 mA, I_Z = 5.78 mA, P_D ≈ 0.96 W; feedback version 13.8 V

4.8 Transistor shunt regulator

Here the transistor sits **across the load** (Fig 4.9b) and acts as an adjustable "dump" for excess current. The series resistor R_S carries the total current $I_S = I_L + I_C$ (+ the small I_B). The Zener sits between output and base, so

SHUNT REGULATOR

$$V_o = V_Z + V_{BE} \qquad I_S = \frac{V_i - V_o}{R_S} \qquad I_C + I_B = I_S - I_L, \quad I_Z = I_B$$

If V_o tries to rise, more current flows through the Zener into the base, the transistor conducts more, more current flows in R_S, and the extra drop pulls V_o back down.

SOLVED EXAMPLE 4.6 · SHUNT REGULATOR

In Fig 4.9b, V_i = 22 V, R_S = 120 Ω, V_Z = 8.2 V, β = 50, R_L = 100 Ω. Find V_o, I_S, I_L, I_C and I_Z.

- V_o = 8.2 + 0.7 = 8.9 V; I_S = (22 − 8.9)/120 = 109.2 mA; I_L = 8.9/100 = 89 mA.
- Shunt current = 109.2 − 89 = 20.2 mA = $I_C + I_B = 51 I_B \rightarrow I_B = I_Z$ = 0.40 mA, I_C = 19.8 mA.

V_o = 8.9 V, I_S = 109 mA, I_L = 89 mA, I_C = 19.8 mA, I_Z = 0.40 mA

Series regulator	Shunt regulator
Control element in series; carries load current	Control element in parallel; carries excess current
Efficient — little wasted at light load	Wasteful — full I_S drawn even at no load
Short circuit destroys pass transistor unless protected	Inherently short-circuit proof (R_S limits current)
Used for most supplies (78xx, LM317)	Used for references, low-current rails (TL431)

4.9 Short-circuit and fold-back protection

If the output of a series regulator is shorted, the pass transistor sees the full V_i across it while carrying a huge current — it dies within milliseconds. The fix is to **sense the load current** with a small resistor R_{sc} and steal base drive when it gets too high (Fig 4.10).

Constant-current limiting

While $I_L R_{sc} < 0.7$ V, Q_2 is off and the regulator works normally. When $I_L R_{sc} = V_{BE2}$, Q_2 conducts, robs Q_1's base current, and the output current is clamped:

$$I_{limit} = \frac{V_{BE2}}{R_{sc}} \quad \text{(e.g. 0.5 A limit} \rightarrow R_{sc} = 0.7/0.5 = 1.4\ \Omega)$$

Problem: under a dead short, Q_1 still carries I_{limit} with the whole V_i across it — $P_D = V_i I_{limit}$ is at its worst.

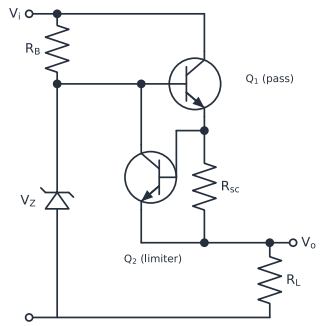


Fig 4.10 Current limiting: Q_2 turns on when $I_L R_{sc}$ reaches V_{BE} and diverts Q_1's base current.

Fold-back limiting

Feed Q_2's base from a divider (ratio $k = R_5/(R_4 + R_5)$) connected from Q_1's emitter to ground, instead of directly. Now Q_2 turns on when $k(V_o + I_L R_{sc}) - V_o = V_{BE}$. As V_o collapses, the turn-on current gets smaller — the current "folds back" (Fig 4.11):

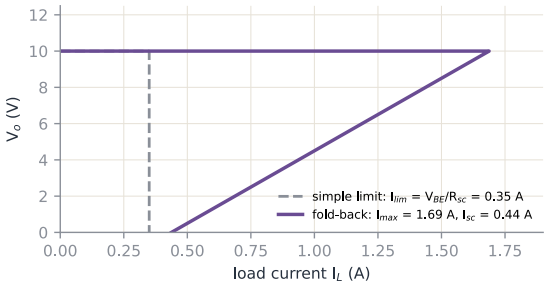


Fig 4.11 Output characteristics: simple limiting vs fold-back (Example 4.7).

FOLD-BACK

$$I_{max} = \frac{V_{BE} + (1-k)V_o}{k\,R_{sc}} \text{ (at rated } V_o) \qquad I_{sc} = \frac{V_{BE}}{k\,R_{sc}} \text{ (at } V_o = 0) \qquad I_{sc} \ll I_{max}$$

SOLVED EXAMPLE 4.7 · FOLD-BACK

A 10 V regulator has $R_{sc} = 2\ \Omega$. (a) With simple limiting, what is I_{limit}? (b) With fold-back, $k = 0.8$: find I_{max} and I_{sc}. (c) With $V_i = 15$ V, compare the pass-transistor dissipation under short circuit.

- (a) $I_{limit} = 0.7/2 = 0.35$ A.
- (b) $I_{max} = (0.7 + 0.2 \times 10)/(0.8 \times 2) = 2.7/1.6 = 1.69$ A; $I_{sc} = 0.7/1.6 = 0.44$ A.

▸ (c) Fold-back allows 1.69 A in normal use but only 0.44 A into a short: $P_D \approx 15 \times 0.44 = 6.6$ W, versus $15 \times 1.69 = 25$ W if a simple limiter were set at 1.69 A.

(a) 0.35 A (b) I_{max} = 1.69 A, I_{sc} = 0.44 A (c) ≈ 6.6 W vs 25 W

4.10 Current boosting with a pass transistor

A three-terminal IC regulator (7805, 7812 ...) is a complete series regulator with built-in protection, but typically good for only ~1 A. To supply more, add an external **PNP pass transistor** (Fig 4.12).

At light load the IC current flows through R_x; while $I_{reg}R_x$ < 0.7 V the PNP is off and the IC supplies everything. Once $I_{reg}R_x$ reaches V_{EB}, the PNP turns on and carries the extra load current from V_i straight to the output. The IC still controls V_o (it sets the PNP's base current), so regulation is preserved.

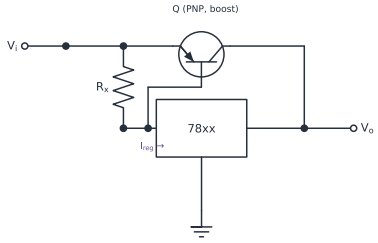


Fig 4.12 Current boosting a 78xx regulator with an external PNP transistor.

CURRENT BOOSTING

$$\text{PNP turns on at } I_{reg} = \frac{V_{EB}}{R_x} \qquad I_L = I_{reg} + I_C, \quad I_{reg} = \frac{V_{EB}}{R_x} + I_B, \quad I_C = \beta I_B$$

SOLVED EXAMPLE 4.8 · CURRENT BOOST

A 7805 is boosted with a PNP (β = 20) and R_x = 7 Ω. At what IC current does the PNP turn on? For I_L = 1 A, find I_C and the IC's own current.

▸ Turn-on: I = 0.7/7 = 100 mA.

▸ $I_L = 0.1 + I_B + 20I_B = 1$ A → I_B = 0.9/21 = 42.9 mA.

▸ $I_C = 20 \times 42.9 = 857$ mA; $I_{reg} = 100 + 42.9 = 143$ mA — the IC is loafing.

Turn-on at 100 mA; I_C = 857 mA, I_{reg} = 143 mA

INTERVIEW CORNER

Q. Why is a linear regulator inefficient, and when do you still choose it?

A. It drops ($V_i - V_o$) at full load current as heat: $\eta \approx V_o/V_i$. You still pick it for low noise, zero switching ripple, simplicity and small $V_i - V_o$ (LDOs) — e.g. powering ADCs and RF sections.

✓ Module 4 — Quick revision

Classes: A 360°, B 180°, AB 180–360°, C < 180°.

η: P_{ac}/P_{dc}; $P_D = P_{dc} - P_{ac}$.

Class A series-fed: η_{max} 25 %. **Transformer-coupled:** 50 %.

Transformer: $R_L' = (N_1/N_2)^2 R_L$; DC load line vertical; swing up to $2V_{CC}$.

Class A max P_D at zero signal = $V_{CC}I_{CQ}$.

Class B: $P_{dc} = 2V_{CC}I_p/\pi$, $P_{ac} = V_p^2/2R_L$, $\eta = \pi V_p/4V_{CC} \le 78.5$ %.

Class B $P_{D(max)}$ = $2V_{CC}^2/\pi^2 R_L$ at $V_p = 2V_{CC}/\pi$.

Crossover distortion: ±0.7 V dead zone in class B; cured by AB bias (diodes / V_{BE} multiplier).

Zener regulator: $I_Z = I_S - I_L$; $R_{S(max)} = (V_{i(min)} - V_Z)/(I_{L(max)} + I_{Z(min)})$.

Series regulator: $V_o = V_Z - V_{BE}$; feedback: $V_o = (1 + R_1/R_2)(V_Z + V_{BE})$.

Shunt regulator: $V_o = V_Z + V_{BE}$; short-circuit proof, inefficient.

Current limit: $I_{lim} = V_{BE}/R_{sc}$. **Fold-back:** $I_{sc} = V_{BE}/kR_{sc} < I_{max}$.

Current boost: PNP on when $I_{reg} = V_{EB}/R_x$; $I_L = I_{reg} + \beta I_B$.

Practice — short questions (3 marks)

1. Classify power amplifiers based on conduction angle with waveforms.
2. Why is a transformer used in a class A power amplifier?
3. What is crossover distortion? How is it eliminated?
4. A class B amplifier with V_{CC} = 20 V drives 16 Ω. Find maximum P_{ac} and maximum dissipation per transistor. [12.5 W; 2.53 W]
5. Define line regulation and load regulation.
6. Compare series and shunt regulators.
7. What is fold-back limiting? Sketch its V–I characteristic.

Practice — long questions (9/14 marks)

1. Derive the maximum efficiency of a transformer-coupled class A amplifier. Draw the DC and AC load lines.
2. Explain class B push-pull operation and derive its maximum efficiency and maximum transistor dissipation.
3. With a neat diagram explain a complementary-symmetry class AB amplifier. Why are diodes used for biasing?
4. Design a Zener shunt regulator for V_o = 9.1 V, V_i = 15–20 V, I_L = 0–30 mA, $I_{Z(min)}$ = 5 mA. [$R_S \le 169$ Ω; with 160 Ω, $P_Z \approx 0.62$ W]
5. Explain a series voltage regulator with feedback and derive its output voltage.
6. Explain short-circuit protection and fold-back limiting in a series regulator. How is current boosting achieved in an IC regulator?

⚡ Placement & GATE rapid-fire

Read a question, answer it aloud in one breath, then check. The ones marked Tricky are favourites for catching people out.

Q. g_m and r_e of a BJT at I_C = 1 mA?
A. 40 mS and 25 Ω (V_T = 25 mV). At 2 mA, g_m = 80 mS.

Q. Does g_m of a BJT depend on β?
A. No — $g_m = I_C/V_T$ only. $r_\pi = \beta/g_m$ does depend on β.

Q. Phase shift of CE, CB, CC stages?
A. CE 180°; CB 0°; CC (emitter follower) 0°.

Q. Which configuration has the highest input resistance? Lowest output?
A. CC (emitter follower) for both — high R_{in}, low R_{out}, gain ≈ 1.

Q. Why is an emitter follower's gain slightly below 1?
A. It is voltage-series feedback with β = 1: $A_f = A/(1 + A)$.

Q. Purpose of a coupling capacitor? Of a bypass capacitor?
A. Coupling: pass AC, block DC between stages. Bypass: short R_E for AC to restore gain while keeping DC stability.

Q. Tricky Remove C_E from a CE amplifier — gain, R_{in}, bandwidth?
A. Gain drops to ≈ $-R_C'/R_E$; R_{in} rises; bandwidth rises (negative feedback).

Q. Which capacitor usually sets f_L of a CE stage?
A. The bypass capacitor C_E, because it sees a tiny resistance (≈ r_e-level).

Q. 2 pF from base to collector, gain −99. Input Miller capacitance?
A. 2 × (1 + 99) = 200 pF.

Q. Define f_T. f_T = 300 MHz, β = 150: f_β?
A. Frequency where short-circuit $|h_{fe}| = 1$. $f_\beta = f_T/\beta$ = 2 MHz.

Q. Two identical stages, each f_H = 1 MHz. Overall f_H?
A. 1 MHz × $\sqrt{(\sqrt{2} - 1)}$ = 0.644 MHz.

Q. Cascode vs cascade?
A. Cascade = stages in series. Cascode = CE/CS stacked with CB/CG in one stage: Miller effect gone, very high R_{out}.

Q. Why are big drain resistors avoided in ICs?
A. Area and DC headroom. A current-source load gives large AC resistance with small DC drop.

Q. MOSFET intrinsic gain with I_D = 0.5 mA, V_{ov} = 0.2 V, λ = 0.02 V^{-1}?
A. $g_m r_o = 2/(\lambda V_{ov}) = 2/(0.02 \times 0.2) = 500$.

Q. Small-signal resistance of a diode-connected MOSFET?
A. ≈ $1/g_m$ (in parallel with r_o).

Q. Tricky Can an enhancement NMOS be self-biased (gate grounded, R_S)?
A. No — $V_{GS} = -I_D R_S \le 0 < V_T$, it never turns on.

Q. Stability factor of fixed bias with β = 100?
A. S = 1 + β = 101 (worst). Voltage-divider bias gives S ≈ 1 + R_{Th}/R_E.

Q. What is thermal runaway?
A. I_C ↑ → heat ↑ → I_{CO}, β ↑ → I_C ↑ … until destruction. Broken by emitter degeneration and heat sinks.

Q. RC high-pass with RC ≪ T fed a square wave — output?
A. Positive and negative spikes at the edges (differentiation).

Q. Rise time of an amplifier with f_H = 3.5 MHz?
A. $t_r = 0.35/f_H$ = 100 ns.

Q. Clamper with a ±5 V square input — output peak-to-peak?
A. Still 10 V. Clampers shift DC level, never change peak-to-peak.

Q. Four things negative feedback improves, and what it costs?
A. Gain stability, bandwidth, distortion, internal noise (and impedances). Cost: gain falls by (1 + Aβ).

Q. Topology of an op-amp inverting amplifier?
A. Voltage-shunt (shunt-shunt); gain $-R_f/R_1$ = transresistance / source resistance.

Q. Topology of a CE stage with unbypassed R_E?
A. Current-series: R_E samples I_E and puts a voltage in series with the input loop.

Q. Tricky Does negative feedback improve signal-to-noise ratio for noise at the input?
A. No. Noise entering with the signal is treated exactly like signal; only noise generated inside the loop is reduced.

Q. Barkhausen criterion?
A. |Aβ| = 1 and total loop phase 0°/360°. Design |Aβ| slightly > 1 so oscillation starts.

Q. What sets an oscillator's amplitude?
A. Non-linearity (saturation) or AGC elements (thermistor, lamp, diodes) that pull $|A\beta|$ back to exactly 1.

Q. Why three RC sections in a phase-shift oscillator?
A. Two sections reach 180° only at infinite frequency; three give 180° at a finite f with $\beta = 1/29$.

Q. RC phase-shift oscillator with R = 1 kΩ, C = 0.1 µF: f_0?
A. $1/(2\pi RC\sqrt{6}) = 650$ Hz; needs $|A| \geq 29$.

Q. Wien bridge with R = 1 kΩ, C = 0.1 µF: f_0 and gain?
A. $1/2\pi RC = 1.59$ kHz; $A \geq 3$ ($R_f \geq 2R_1$).

Q. Colpitts with L = 10 µH, $C_1 = C_2 = 200$ pF?
A. $C_{eq} = 100$ pF → $f_0 = 1/(2\pi\sqrt{(10\ \mu \times 100\ p)}) \approx 5.03$ MHz.

Q. Which oscillator for 1 kHz audio, 10 MHz RF, and a watch?
A. Wien bridge; Colpitts/Hartley; 32.768 kHz crystal (Pierce).

Q. Why is a crystal oscillator so stable?
A. Very high Q and quartz L, C barely change with temperature; a tiny frequency shift causes a big phase change, so the loop locks tightly.

Q. Where between f_s and f_p is a crystal inductive?
A. Only between them — that is where a Pierce/Colpitts crystal oscillator runs.

Q. Max efficiency of class A (resistive), class A (transformer), class B?
A. 25 %, 50 %, 78.5 % ($\pi/4$).

Q. Tricky **Class B efficiency at half output swing ($V_p = V_{CC}/2$)?**
A. $\eta = (\pi/4)(1/2) = 39.3$ %. Efficiency is proportional to V_p.

Q. When does a class A transistor dissipate most?
A. With no signal — all of P_{dc} becomes heat.

Q. When does a class B transistor dissipate most?
A. At $V_p = 2V_{CC}/\pi \approx 0.64V_{CC}$; $P_{D(max)} = 2V_{CC}^2/\pi^2R_L$ for both.

Q. Cause and cure of crossover distortion?
A. The 0.7 V V_{BE} dead zone in class B. Cure: class AB bias with diodes or a V_{BE} multiplier.

Q. Why does push-pull cancel even harmonics?
A. The two halves are mirror images; even-order terms appear with the same sign in both collectors and subtract in the output transformer.

Q. Where is class C used and why can't it be used for audio?
A. RF transmitters with a tuned tank; the current is a pulse train, so wide-band (audio) signals would be badly distorted.

Q. Zener regulator: what two conditions fix R_S?
A. Minimum V_i with maximum load must still give $I_{Z(min)}$; maximum V_i with no load must keep P_Z under its rating.

Q. Series vs shunt regulator — which survives a short circuit by itself?
A. Shunt (R_S limits current). A series pass transistor needs current limiting.

Q. What is dropout voltage?
A. Minimum $V_i - V_o$ for regulation: ≈ 2 V for a 78xx, 0.1–0.3 V for an LDO.

Q. Why fold-back rather than simple current limiting?
A. Under a short it cuts current to a small I_{sc}, keeping pass-transistor dissipation low, while still allowing high current normally.

Q. Efficiency of a linear regulator dropping 12 V to 5 V?
A. ≈ $V_o/V_i = 5/12 \approx 42$ % — the rest is heat. That is why switching regulators exist.

Q. How does an external PNP boost a 7805?
A. When $I_{reg}R_x$ reaches 0.7 V the PNP turns on and carries the extra current; the IC still sets V_o.

∑ Formula sheet

Module 1 — Wave shaping & biasing

Quantity	Formula
RC step responses	HPF: $V\,e^{-t/RC}$ · LPF: $V(1 - e^{-t/RC})$
Square wave into HPF	$V_1 = V/(1 + e^{-T/2RC})$, $V_1' = V_1 e^{-T/2RC}$, tilt ≈ $T/2RC = \pi f_1/f$
Pulse tilt · rise time	$P \approx t_p/RC \cdot t_r = 2.2RC = 0.35/f_2$
First-order filters	$f_c = 1/2\pi RC$; $\lvert H_{LP}\rvert = 1/\sqrt{(1 + (f/f_c)^2)}$; $\lvert H_{HP}\rvert = 1/\sqrt{(1 + (f_c/f)^2)}$
Clipper / clamper levels	Biased shunt: $V_R + V_\gamma$ · Clamper: shift = $V_m - V_\gamma$; needs $R_L C \gg T/2$
Stability factor	$S = (1 + \beta)/(1 - \beta\,\partial I_B/\partial I_C)$
Fixed bias	$I_B = (V_{CC} - V_{BE})/R_B$; $S = 1 + \beta$
Collector-to-base	$I_B = (V_{CC} - V_{BE})/(R_B + (1 + \beta)R_C)$; $S = (1 + \beta)/(1 + \beta R_C/(R_C + R_B))$
Voltage divider	$I_B = (V_{Th} - V_{BE})/(R_{Th} + (1 + \beta)R_E)$; $S = (1 + \beta)(1 + R_{Th}/R_E)/(1 + \beta + R_{Th}/R_E)$
MOSFET bias	$I_D = K(V_{GS} - V_T)^2$; $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$; self: $V_{GS} = -I_D R_S$; drain-fb: $V_{GS} = V_{DS}$

Module 2 — Amplifiers

Quantity	Formula
Hybrid-π	$g_m = I_C/V_T$; $r_\pi = \beta/g_m$; $r_o = V_A/I_C$
CE mid-band	$A_v = -g_m(r_o \parallel R_C \parallel R_L)$; $R_{in} = R_B \parallel r_\pi$; $R_{out} = r_o \parallel R_C$; $A_{vs} = A_v R_{in}/(R_{in} + R_s)$
Low-f poles	$1/2\pi(R_s + R_{in})C_{C1}$; $1/2\pi(R_{out} + R_L)C_{C2}$; $1/2\pi R_e' C_E$, $R_e' = R_E \parallel (r_\pi + R_s \parallel R_B)/(\beta + 1)$
Miller	$C_{in} = C(1 + A)$; $C_{out} = C(1 + 1/A)$; $f_H = 1/2\pi R_{sig}'(C_\pi + C_\mu(1 + g_m R_L'))$
f_β, f_T	$f_\beta = 1/2\pi r_\pi(C_\pi + C_\mu)$; $f_T = \beta f_\beta = g_m/2\pi(C_\pi + C_\mu)$
MOSFET	$g_m = 2K\,V_{ov} = 2I_D/V_{ov} = 2\sqrt{(KI_D)}$; $r_o = 1/\lambda I_D$
CS loads	$-g_m(R_D \parallel r_o)$; $-g_m(r_{o1} \parallel r_{o2})$; $-g_{m1}/g_{m2} = -\sqrt{((W/L)_1/(W/L)_2)}$
Cascade	$A = \Pi A_k$; $f_H(n) = f_H\sqrt{(2^{1/n} - 1)}$; $f_L(n) = f_L/\sqrt{(2^{1/n} - 1)}$
Cascode	$A_v \approx -g_{m1}R_{load}$; $R_{out} \approx \beta r_o$ (BJT), $g_{m2}r_{o2}r_{o1}$ (MOS)

Module 3 — Feedback & oscillators

Quantity	Formula
Closed loop	$A_f = A/(1 + A\beta)$; $dA_f/A_f = (dA/A)/(1 + A\beta)$
Bandwidth, distortion	$f_{Hf} = f_H(1 + A\beta)$; $f_{Lf} = f_L/(1 + A\beta)$; $D_f = D/(1 + A\beta)$
Impedances	Series mixing $R_i(1 + A\beta)$; shunt $R_i/(1 + A\beta)$; voltage sampling $R_o/(1 + A\beta)$; current sampling $R_o(1 + A\beta)$
Voltage-shunt	$\beta = -1/R_F$; $R_{mf} = R_m/(1 + \beta R_m)$; $A_{vf} = R_{mf}/R_s \to -R_F/R_s$
RC phase-shift	$f = 1/(2\pi RC\sqrt{6})$; $\beta = -1/29$; $\lvert A\rvert \geq 29$
Wien bridge	$f = 1/2\pi RC$; $\beta = 1/3$; $A \geq 3$
Hartley / Colpitts	$f = 1/2\pi\sqrt{((L_1 + L_2 + 2M)C)}$, $A \geq L_1/L_2$ · $f = 1/2\pi\sqrt{(LC_1C_2/(C_1 + C_2))}$, $A \geq C_2/C_1$
Crystal	$f_s = 1/2\pi\sqrt{(LC)}$; $f_p = 1/2\pi\sqrt{(LCC_M/(C + C_M))} = f_s\sqrt{(1 + C/C_M)}$

Module 4 — Power amplifiers & regulators

Quantity	Formula
Class A transformer	$R_L' = n^2 R_L$; $I_{CQ} = V_{CC}/R_L'$; $P_{dc} = V_{CC}I_{CQ}$; $P_{ac} = V_p^2/2R_L'$; $\eta_{max} = 50\ \%$
Class B	$P_{dc} = 2V_{CC}I_p/\pi$; $P_{ac} = V_p^2/2R_L$; $\eta = \pi V_p/4V_{CC} \leq 78.5\ \%$
Class B dissipation	$P_{D(max)} = 2V_{CC}^2/\pi^2 R_L$ (both) at $V_p = 2V_{CC}/\pi$
Zener regulator	$I_Z = (V_i - V_Z)/R_S - V_Z/R_L$; $R_{S(max)} = (V_{i(min)} - V_Z)/(I_{L(max)} + I_{Z(min)})$
Series / shunt	$V_o = V_Z - V_{BE}$ · feedback $V_o = (1 + R_1/R_2)(V_Z + V_{BE})$ · shunt $V_o = V_Z + V_{BE}$
Protection	$I_{lim} = V_{BE}/R_{sc}$; fold-back $I_{sc} = V_{BE}/kR_{sc}$, $I_{max} = (V_{BE} + (1 - k)V_o)/kR_{sc}$
Current boost	PNP on at $I_{reg} = V_{EB}/R_x$; $I_L = I_{reg} + \beta I_B$
Regulation	Line = $\Delta V_o/\Delta V_i$; load = $(V_{NL} - V_{FL})/V_{FL} \times 100\ \%$

☾ Night before the exam

You have done the work. Tonight is for calm, quick recall — not new topics. Go through this page once, sleep well.

M1 must-draws: square wave through HPF/LPF for RC ≪ T, ≈ T, ≫ T; clipper transfer curve; clamper output; load line with Q.

M1 must-derive: V_1, V_1' and tilt; integrator/ differentiator conditions; S for collector-to-base and voltage-divider bias.

M1 numericals: VDB exact (Thevenin) and approximate; MOSFET quadratic — always reject the root with $V_{GS} < V_T$.

M2 must-draws: RC-coupled CE amp, hybrid-π, mid-band equivalent, frequency response with three regions, cascode.

M2 must-derive: A_v, R_{in}, R_{out}; Miller capacitance; $h_{fe}(f)$, f_β, f_T; bandwidth shrinkage $\sqrt{(2^{1/n} - 1)}$.

M2 numericals: g_m = 40 mS/mA; C_E sets f_L; $C_M = C_\mu(1 + g_m R_L')$.

M3 must-draws: feedback block diagram; four topologies; RC phase-shift, Wien, Hartley, Colpitts; crystal equivalent and reactance curve.

M3 must-derive: A_f, desensitivity, f_{Hf}; R_{if}, R_{of} for voltage-series; β = −1/29 at $1/(2\pi RC\sqrt{6})$; β = 1/3 at $1/2\pi RC$.

M3 one-liners: series mixing ↑ R_{in}, shunt ↓ R_{in}; voltage sampling ↓ R_{out}, current ↑ R_{out}.

M4 must-draws: conduction-angle waveforms; transformer-coupled class A with load lines; push-pull and complementary symmetry; crossover distortion; Zener, series, shunt regulators; fold-back curve.

M4 must-derive: class A η = 50 %; class B η = π/4, $P_{D(max)}$ at $V_p = 2V_{CC}/\pi$.

M4 numericals: check whether the Zener is ON first; design R_S at $V_{i(min)}$, $I_{L(max)}$; check P_Z at $V_{i(max)}$, no load.

Exam-hall habits that win marks

1. Start with the question you are most sure of — confidence compounds.
2. For every derivation: neat diagram first, label every current and voltage, then equations.
3. Write the formula before substituting numbers; examiners give method marks even if arithmetic slips.
4. Box the final answer with its unit (mA, kΩ, Hz, %). A missing unit can cost a mark.
5. Sanity-check: is V_{CE} positive? Is η below its class limit? Is the Zener actually on? Is $f_H > f_L$?
6. Keep 10 minutes at the end to re-read and complete any half-drawn waveform.
7. Remember the four big ideas: a capacitor resists sudden voltage change, a transistor is a controlled current source, feedback trades gain for quality, and power is V × I. All the best!